

Online Appendix

for

Who's Afraid of the Minimum Wage? Measuring the Impacts on Independent Businesses Using Matched U.S. Tax Returns

by

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This appendix includes several sections of supplementary information and results.

- Appendix [A](#) describes the details of the minimum wage increases we study.
- Appendix [B](#) explains how we screen for highly exposed industries using public data from the Current Population Survey.
- Appendix [C](#) provides additional summary statistics.
- Appendix [D](#) compares our own-wage elasticities estimates to estimates from previous studies.
- Appendix [E](#) reports detailed estimates for all firm outcomes and impacts on firm deductions outside of wage bills and materials costs.
- Appendix [F](#) examines the heterogeneity of firm responses.
- Appendix [G](#) breaks down firm responses by industry.
- Appendix [H](#) discusses the robustness of the firm results.
- Appendix [I](#) examines the robustness of the individual results.
- Appendix [J](#) reports alternative firm and individual estimates using regression adjusted estimators.
- Appendix [K](#) presents additional worker transition analysis.
- Appendix [L](#) details the sources and construction of our IRS data samples.
- Appendix [M](#) reports estimates from an alternative border design.
- Appendix [N](#) presents results from a parallel analysis of individual earnings and employment responses using public ACS data.
- Appendix [O](#) presents a theoretical framework consistent with our empirical results.

A. Minimum Wage Changes and Control States

Our analysis examines the impacts of the 19 state and city minimum wage increases that began between 2013 and 2016 and did not exempt small firms. Along with 17 state-level changes, we also include Chicago’s minimum wage increase from 2015 and Washington D.C.’s increase from 2014. We exclude Arizona’s 2017 \$1.95 increase in its minimum wage as the Arizona wage floor does not apply to firms with revenues less than \$500K.

Our set of control states comprises all states that did not adjust their minimum wages between 2011 and 2019. Appendix Table A.1 lists the treatment and control states used in our analysis along with details of the size of each minimum wage increase. Note, Illinois and Nevada had small minium wage increases in the first year of our sample period, 2010, of 25 cents and 70 cents respectively. In our regression specifications, we include an indicator variable for these state-years, meaning that these control states do not contribute to the counterfactual for event-year $s - 4$ for the 2013 cohort of minimum wage increases. The timing and details of the policy changes are further described in Appendix Table A.2 below.

Table A.1: States used for Minimum Wage Analyses

Treatment				Control
state	cohort	MW increase (\$)	MW increase (%)	state
NY	2013	\$2.45	33.8%	AL
RI	2013	\$2.20	29.7%	GA
CA	2014	\$3.00	37.5%	ID
CT	2014	\$1.85	22.4%	IL (non-Chi)
DC	2014	\$5.00	60.6%	IN
DE	2014	\$1.00	13.8%	KS
MI	2014	\$1.85	25.0%	LA
MN	2014	\$2.40	33.1%	MS
NJ	2014	\$1.35	18.6%	NC
AK	2015	\$2.14	29.5%	ND
AR	2015	\$2.00	27.6%	NH
HI	2015	\$2.85	39.3%	NV
Chicago,IL	2015	\$4.75	57.6%	OK
MA	2015	\$4.00	50.0%	PA
MD	2015	\$3.75	51.7%	SC
NE	2015	\$1.75	24.1%	TN
SD	2015	\$1.85	25.5%	TX
WV	2015	\$1.50	20.7%	UT
ME	2016	\$3.50	46.7%	VA
avg (firm weighted)		\$2.64	34.6%	WI
avg (unweighted)		\$2.59	34.1%	WY

Note: The table above lists the treatment states and localities with minimum wage increases and the control states used in the analysis. Each minimum wage increase is grouped into a cohort based on the initial year of the increase. The table details the size of the increase four years after the policy change ($s + 4$) in dollar terms and as a percentage increase relative to the year prior to the minimum wage increases ($s - 1$).

Table A.2: Details of Minimum Wage Increases Used in Analysis

State	Enactment	Effective	Enacted MW	Prior MW	Phase-In	Exclusions	Next MW Change	
							Date	Details
NY	03/29/13	12/31/13	\$8.00	\$7.25	\$8.75 on 12/31/14 and \$9.00 on 12/31/15	–	12/31/16	MW in NYC for firms with more than 10 employees increases to \$11.00 on 12/31/16, to \$13.00 on 12/31/17, to \$15.00 on 12/31/18. MW increases were lower outside NYC, and fast food workers were subject to higher MWs.
RI	06/11/12	01/01/13	\$7.75	\$7.40	–	–	09/03/17	MW increases to \$10.10 on 01/01/18 and to \$10.50 on 01/01/19
	07/16/13	01/01/14	\$8.00	\$7.75				
	07/03/14	01/01/15	\$9.00	\$8.00				
	06/22/15	01/01/16	\$9.60	\$9.00				
CA	09/25/13	07/01/14	\$9.00	\$8.00	\$10.00 on 01/01/16	–	04/04/16	MW increases to \$10.50 on 01/01/17 and \$11 on 01/01/18 for employers with 26+ employees, then increases by \$1.00 a year through 2022 when it reaches \$15 and is then inflation indexed; employers with 25 or fewer employees have a 1-year delay for all increases

CT	06/06/13	01/01/14	\$8.70	\$8.25	\$9.00 on 01/01/15	–	03/27/14	MW increases to \$9.15 on 01/01/15, then to \$9.60 on 01/01/16, and finally to \$10.10 on 01/01/17
DC	01/15/14	07/01/14	\$9.50	\$8.25	\$10.50 on 07/01/15 and \$11.50 07/01/16	–	06/27/16	MW increases to \$12.50 on 07/01/17 and gradually increase to \$15.00 in 2020
DE	01/30/14	06/01/14	\$7.75	\$7.25	\$8.25 on 06/01/15	–	07/01/18	MW increases to \$8.25 on 10/01/18, to \$9.25 on 10/01/19, to \$9.75 on 10/01/20 and then to \$10.25 on 10/01/21
MI	05/27/14	09/01/14	\$8.15	\$7.40	\$8.50 on 01/01/16, \$8.90 on 01/01/17 and \$9.25 on 01/01/18	Firms with only a single employee	12/04/18	MW increases to \$9.45 on 03/29/19, then potentially rises to \$12 in 2030
MN	04/14/14	08/01/14	\$8.00	\$7.25	\$8.00 on 08/01/14, \$9.00 on 08/01/15 and \$9.50 on 8/01/16	Firms with gross revenues less than \$500K required to pay roughly 80% of standard minimum wage	12/17/17	MW increases with cost of living beginning on 01/01/18
NJ	11/05/13	01/01/14	\$8.25	\$7.25	Indexed to inflation	–	02/04/19	MW increases to \$10.00 on 07/01/19, \$11.00 on 01/01/20, with \$1.00 increases until 01/01/24 when MW will be 2024

AK	11/04/14	01/01/15	\$8.75	\$7.75	\$9.75 on 01/01/16, and increased by cost of living or to \$1.00 above the federal minimum wage if it is higher thereafter.	—	—	—
AR	11/06/14	01/01/15	\$7.50	\$7.25	\$8.00 on 01/01/16 and \$8.50 on 01/01/17	Firms with fewer than four employees	11/04/18	MW increases to \$9.25 on 01/01/19, to \$10.00 on 01/01/20, and to \$11.00 on 01/01/21
HI	05/23/14	01/01/15	\$7.75	\$7.25	\$8.50 on 01/01/16, \$9.25 on 01/01/17 and \$10.10 on 01/01/18	—	09/29/22	MW increases to \$12.00 on 10/01/22, to \$14.00 on 01/01/24, to \$16.00 on 01/01/26, and to \$18.00 on 01/01/28
IL-Chicago	12/02/14	07/01/15	\$10.00	\$8.25	\$10.50 on 07/01/16, \$11.00 on 07/01/17, \$12.00 on 07/01/18, and \$13.00 on 07/01/19	—	06/06/23	MW increases to \$15.80 for employers with 21 or more employees and \$15.00 for employers with 4 to 20 employees on 07/01/23
MA	07/06/14	01/01/15	\$9.00	\$8.00	\$10.00 on 01/01/16, \$11.00 on 01/01/17	—	06/29/18	MW increases to \$12.00 on 01/01/19, \$12.75 on 01/01/20, \$13.50 on 01/01/21, \$14.25 on 01/01/22, \$15.00 on 01/01/23
MD	05/05/14	01/01/15	\$8.00	\$7.25	\$8.25 on 07/01/15, \$8.75 on 07/01/16, \$9.25 on 07/01/17 and \$10.10 on 07/01/18	—	03/28/19	MW increases to \$11.00 on 01/01/20, to \$11.75 on 01/01/21, to \$12.50 on 01/01/22, to \$13.25 on 01/01/23, to \$14.00 on 01/01/24

NE	11/04/14	01/01/15	\$8.00	\$7.25	\$9.00 on 01/01/16	–	11/03/20	MW increases to \$10.00 on 01/01/23, to \$12.00 on 01/01/24, to \$13.50 on 01/01/24, to \$15.00 on 01/01/26, then increases with cost of living beginning on 01/01/27
SD	11/04/14	01/01/17	\$8.50	\$7.25	Increased by cost of living thereafter	–	–	–
WV	02/05/14	01/01/15	\$8.00	\$7.25	\$8.75 on 01/01/16	–	–	–
ME	11/08/16	01/01/17	\$9.00	\$7.50	\$10.00 on 01/01/18, \$11.00 on 01/01/19, \$12.00 on 01/01/20, and increased by cost of living beginning on 01/01/21	–	–	–

Note: The table above details the minimum wage increases used to identify our estimates. The first column lists the enactment date and the second column lists the date the policy went into effect. The third and fourth columns show the new minimum wage and the prevailing prior minimum wage. The next column describes any phase-in of minimum wages legislated on the enactment date. The next column describes any exclusions and the final columns detail any further minimum wage policies enacted subsequent to the studied policy.

B. Screening Industries Using CPS Data

While the IRS data comprehensively describe the revenues, costs and profits of firms and the income histories of their workers, tax data do not include wage measures. Wage data are necessary to identify the industries in which minimum wage workers are most prevalent — and thus where minimum wage increases will have discernible impacts.

To screen for industries with relatively high shares of minimum wage workers, we turn to public Current Population Survey (CPS) data, which records the wage rates of surveyed workers. Specifically, we use the Monthly Outgoing Rotation Groups of the CPS to measure the share of minimum wage workers in each four-digit industry. We focus on treatment states and localities in the year before each jurisdiction’s minimum wage increase to measure which industries were most likely to be impacted by the legislation. For each industry i in state or locality j that raised its minimum wage in event time s , we calculate:

$$share_{is-1} = \frac{\sum_j n_{ijs-1}}{\sum_i \sum_j n_{ijs-1}} \quad (\text{B.1})$$

where n_{ijs-1} is the number of non-allocated (non-imputed) workers employed in industry i in state j paid the prevailing state minimum wage or less in the year before the state or locality raises its minimum wage. The numerator sums across jurisdictions to determine the total number of minimum wage workers in the industry across all treated states and localities in the relevant base years. The denominator reports the aggregate total number of non-allocated (non-imputed) workers paid the minimum wage or less in treated jurisdictions in the base year. It is worth noting that these state-level worker counts come from different calendar years as we aim to measure the share of potentially affected workers in the year before the minimum wage increase, that is in event time $s - 1$.

The CPS uses [Census 2007 industry codes](#) and we calculate the above minimum wage worker shares at the 4-digit Census Industry Code level. The IRS data, on the other hand, characterizes firm industries according to the [2017 NAICS industry classification system](#). As such, we map the minimum wage worker shares we construct from CPS data from Census 2007 industry codes to 4-digit 2017 NAICS codes. To do so we first map the CPS industries codes to the 2012 NAICS codes using this [crosswalk](#), then map 2012 NAICS codes to the 2017 NAICS codes used by the IRS. Our analysis includes all industries where the share of minimum wage workers in the industry exceeds 1%, which are listed in Table [B.3](#). We describe these industries as “highly exposed” to minimum wage policies. Collectively they account for more than two-thirds of minimum wage workers.

Table B.3: Industries with at least 1% of minimum wage workers, Current Population Survey (Monthly Outgoing Rotation Groups)

Census 2007	Industry Name	Share of MW Workers	2017 NAICS
290	Support activities for agriculture and forestry	0.01001	1151, 1152, 1153
8690	Drinking places, alcoholic beverages	0.01019	7224
7890	Other schools and instruction, and educational support services	0.01044	6116, 6117
7860	Elementary and secondary schools	0.01129	6111
7870	Colleges and universities, including junior colleges	0.01289	6112, 6113
7690	Services to buildings and dwellings	0.01367	5617
1680	Cut and sew apparel manufacturing	0.01685	3152, 3159
8660	Traveler accommodation	0.02056	7211
5170	Clothing stores	0.02401	4481
5380	Department stores and discount stores	0.02771	4522
4970	Grocery stores	0.04102	4451
8590	Other amusement, gambling, and recreation industries	0.04607	7131, 7132, 7139
8680	Restaurants and other food services	0.42242	7223, 7225
Total		0.66713	

Note: This table lists four-digit NAICS industries that employ at least 1% of workers paid at or below the state minimum wage in the year prior to the minimum wage increase. Tabulations are from the Current Population Survey Monthly Outgoing Rotation Groups (CPS MORGS) and describe the ratio of non-allocated (non-imputed) workers paid at or below the minimum wage in an industry divided by the total non-allocated (non-imputed) number of workers paid at or below the minimum wage in aggregate (across all industries). The state-level worker counts come from different calendar years as the minimum wage increases begin in different years. Because the CPS MORGS uses Census 2007 industry codes while the IRS data uses 2017 NAICS codes we use a [crosswalk](#) to map between the datasets. We do not include employment services in our set of analyzed industries as we do not know whether the industry actually employing these temporary workers is highly exposed. We also exclude crop production due to substantial labor market differences relative to other industries, such as strong seasonality, and because the impact on this industry would be almost solely identified by the California minimum wage increase.

C. Additional Summary Statistics

Appendix Table C.4 reports the share of firms that are pass-throughs by size. The two right-most columns present the overall size distribution of all firms, pass-throughs and non-pass-throughs, as well as the percentage of total employment accounted for by firms within each size range. The first two columns show the percentage of firms and employment within each size range specific to pass-through firms. Summary statistics for the individual panels of low-earning workers and young individuals are presented in Appendix Table C.5.

Table C.4: Pass-Through Businesses and All Businesses by Size

Number of employees	Pass-through shares		Shares of total by size	
	share of firms	share of employment	share of firms	share of employment
0-4	84%	83%	62%	5%
5-9	81%	80%	17%	6%
10-19	78%	78%	10%	7%
20-99	76%	75%	9%	17%
100-499	68%	67%	1%	13%
500+	53%	24%	0.4%	52%
<20	83%	80%	89%	18%
<100	82%	78%	98%	35%
<500	82%	75%	99.6%	48%
Total	82%	49%	100%	100%

Source: Authors' calculations using the *Census County Business Patterns tables by Legal Form of Organization (LFO), Employment Size and NAICS Sector*, the [SOI Integrated Business Data](#) and [Census's SUSM Annual Data Tables by Establishment Industry](#).

Table C.5: Summary Statistics - Individual Panels (base year means)

	Low earning workers panel		Young Individuals panel			
	Low earning all	Low earning exposed	Young all	Young exposed	Teen all	Teen exposed
Earnings (\$)	8,244	7,760	16,153	11,366	5,565	5,561
Age	31.5	27.7	22.28	21.32	18.05	18.02
young share	0.50	0.64	1	1	1	1
teen share	0.16	0.25	0.18	0.29	1	1
N jobs	1.6	1.6	1.6	1.8	1.5	1.6

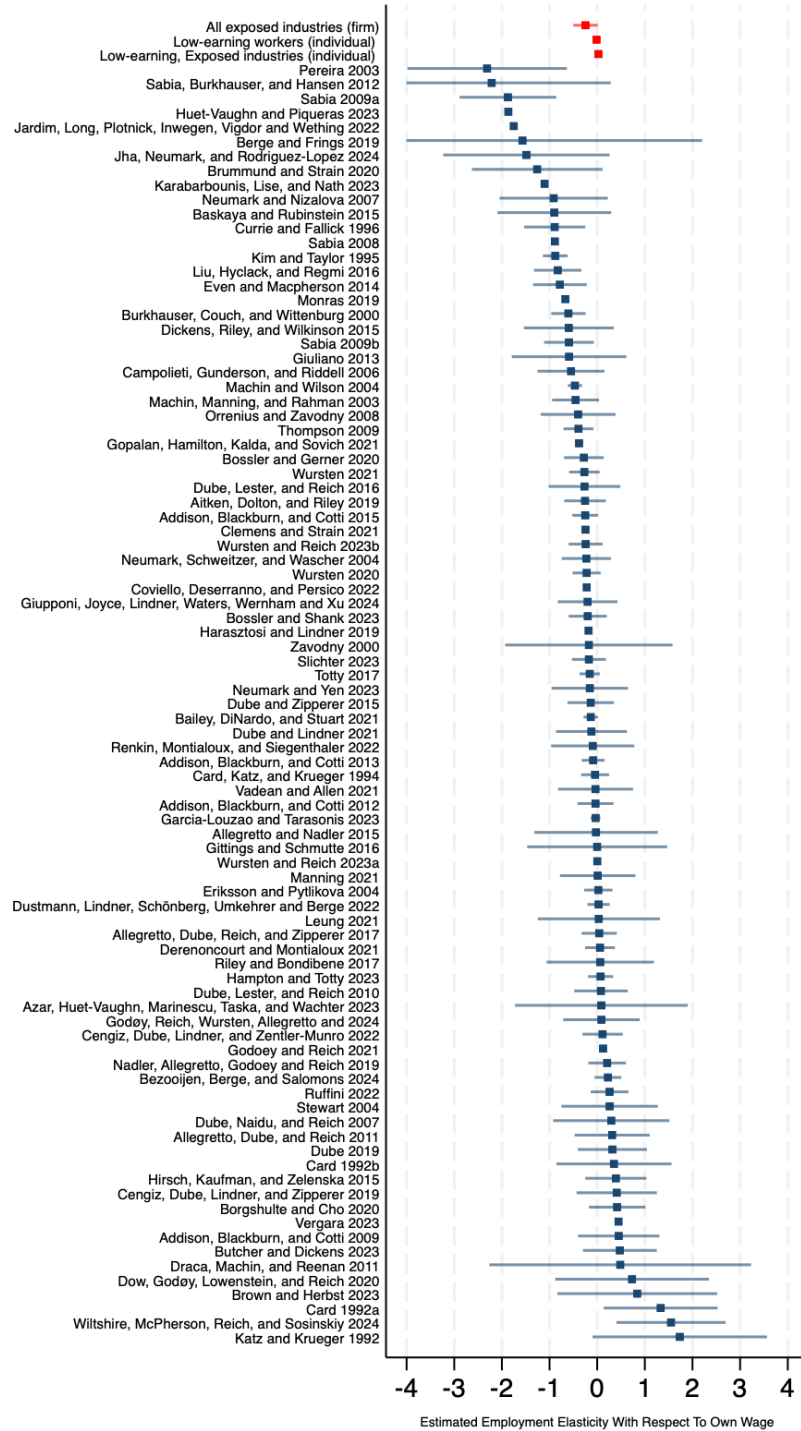
Note: This table presents baseline means for the low-earning worker panel and the panel of young individuals in treatment states in pre-reform year $s - 1$. Means are reported for all industries and highly exposed industries. Earnings represent the sum of wage and salary earnings across all jobs; young and teen shares are the percent of each sample that are 15-26 years old and 15-19 years old, respectively; "N jobs" is the number of distinct employers in the year.

D. Comparison of OWE estimates with the Literature

We generate own-wage elasticity (OWE) estimates from both the firm data describing employment at highly exposed independent businesses, and the individual data tracking low-earning workers. We plot these OWE estimates in red alongside OWE estimates from previous studies in blue.

We estimate firm-level OWEs by separately estimating the change in log employment and the change in log average wages, using Eq. (1), and dividing the change in log employment by the change in log average wages. Individual panel OWEs are analogously calculated as the percent change in employment over the percent change in average annual earnings. For employment, the percentage point changes displayed in Figure 8 are converted to percent changes, evaluated using the observed employment rates among control firms as a counterfactual. The percent change in annual earnings is estimated using Specification (2) where the outcome is log individual annual earnings.

Figure D.1: Employment Elasticity Comparison



Note: This figure compares the own-wage employment elasticities (OWEs) estimated in this study to those from previous studies reported in the *Minimum Wage Own-Wage Elasticity repository* (Dube and Zipperer, 2024). The point estimates and 95% confidence intervals from this study are at the top in red. The top estimate is the OWE from the main sample of independent businesses in highly exposed industries (from the firm-level analysis). The second two are OWE estimates from the low-earning worker panel (individual-level analysis) — one for all baseline low-earning workers and one for low-earning workers employed in highly exposed industries at baseline. Each estimate represents the change in employment four years after the minimum wage increases. The standard errors for each are calculated using the Delta method.

E. Impacts on Other Costs and Detailed Estimates

Appendix Table E.6 reports estimates of the impact of higher state minimum wages on other deductions such as officer compensation or depreciation expenses among pass-through firms in highly exposed industries. While wage bills and COGS rise with the minimum wage as reported in the main text, no other costs are meaningfully impacted by the minimum wage increases, though pension and interest expenses exhibit a statistically significant increase.

Appendix Table E.7 reports four-year estimates of all firm outcomes for the balanced panel of independent businesses in all highly exposed industries and by industry group, as well as for an unbalanced panel.

Table E.6: Effect on Other Income Statement Items

	Outcomes scaled by baseline revenues, $\Delta s-1$ to $s+4$							
	Officer comp	Rent deductions	Tax deductions	Repairs	Pensions	Benefits deductions	Interest	Depreciation
All exposed	-0.0002 (0.0008)	0.0011 (0.0007)	0.0013 (0.0010)	-0.0003 (0.0003)	0.00002** (0.00001)	-0.0001 (0.0001)	0.0009*** (0.0003)	0.0007* (0.0004)
N	530,412	530,412	530,412	530,412	530,412	530,412	530,412	530,412

Note: This table presents estimated effects of the minimum wage increases on independent businesses in highly exposed industries (main sample). Each column represents different firm costs (reported deduction on the firms' tax returns). Depreciation includes Section 179 expensing, which allows for immediate expensing of up to \$500,000 in new investment from 2010-2017, and up to \$1 million in 2018 and 2019, for small and medium sized firms, defined as those with less than \$2 million (2010-2017) or \$2.5 million (2018 and 2019) in new purchases in the year. Therefore, for the majority of firms in this sample, new investment could be expensed immediately as a Section 179 deduction. Each estimate is from a firm-level regression of Equation (1); the coefficients are estimated effects on the outcomes four years after the minimum wage increase, $s + 4$, relative to the base year, $s - 1$. Outcomes are scaled by baseline revenue and are weighted by log baseline revenue. In addition to firm-cohort and cohort-by-year fixed effects, the specifications include categories of baseline firm size (# of workers), deciles of baseline firm value-added, baseline two-digit industry, quintiles of county density, and quintiles of county employment rates – all flexibly interacted with with year to allow for differential time trends. Standard errors are clustered at the state-by-cohort level.

Table E.7: Detailed Firm Outcomes - by Industry and for an Unbalanced Panel

	Outcomes scaled by baseline revenues, $\Delta s-1$ to $s+4$			
	All exposed	Restaurants	Other/retail	Unbalanced (all)
Revenue	0.0331** (0.0149)	0.0294** (0.0128)	0.0359** (0.0178)	0.0377** (0.0179)
Wage bill	0.0143*** (0.0032)	0.0201*** (0.0041)	0.0076*** (0.0022)	0.0128*** (0.0028)
COGS	0.0131*** (0.0038)	0.0053 (0.0036)	0.0210*** (0.0045)	0.0126** (0.0052)
Other deductions	0.0035 (0.0037)	0.0029 (0.0032)	0.0039 (0.0043)	0.0039 (0.0032)
Variable costs	0.0300*** (0.0101)	0.0282*** (0.0100)	0.0310*** (0.0107)	0.0283** (0.0109)
Expensed investment	0.0007* (0.0004)	0.0001 (0.0003)	0.0010 (0.0005)	0.0007 (0.0005)
Value added	0.0221* (0.0113)	0.0245** (0.0096)	0.0184 (0.0134)	0.0259** (0.0120)
Owner Profits	0.0002 (0.0029)	-0.0001 (0.0019)	0.0003 (0.0045)	0.0045 (0.0047)
Employment	-1.496** (0.602)	-2.306** (0.915)	-0.683** (0.318)	-0.435 (0.343)
Own wage elasticity	-0.254* (0.134)	-0.364** (0.138)	-0.0004 (0.1490)	-0.259** (0.136)
N firms	134,974	68,193	66,781	256,117

Note: This table presents estimated effects of the minimum wage increases on independent businesses across subsamples of the highly exposed industries, and for an alternate sample. The first column shows outcomes for all highly exposed industries, Col. (2) for restaurants only, and Col. (3) for other, non-restaurant, highly exposed industries (detailed further in Appendix Tables G.15). The fourth column presents results for an unbalanced sample, consisting of firms active in the pre-reform year $s-1$, that may exit after that year (and may have entered after year $s-4$); in years where the firm is inactive, outcomes are coded as zeros. In the first 8 rows, the outcome is the variable defined in each row scaled by baseline ($s-1$) revenue. The own-wage elasticity (OWE) is calculated as the estimated change in log employment over the estimated change in log average wages. Each estimate is from a firm-level regression of Equation (1); the coefficients are estimated effects on the outcomes four years after the minimum wage increase, $s+4$, relative to the base year, $s-1$. Regressions where outcomes are scaled by baseline revenue are weighted by log baseline revenue; when estimating log changes, regressions are scaled by the baseline value of the outcome. In addition to firm-cohort and cohort-by-year fixed effects, the specifications include categories of baseline firm size (# of workers), deciles of baseline firm value-added, baseline two-digit industry, quintiles of county density, and quintiles of county employment rates – all flexibly interacted with year to allow for differential time trends. Standard errors are clustered at the state-by-cohort level.

F. Heterogeneity of Firm Results

The firm impacts reported in the main text describe how minimum wage increases affect the average firm in a highly exposed industry. This appendix explores the heterogeneity of firm responses by baseline characteristics and treatment. Appendix Tables F.8 through F.10 report estimates by baseline revenue, low-earning labor share of variable costs, and baseline value-added per worker. Appendix Table F.11 examines heterogeneity by treatment year cohort and when excluding California, while Appendix Table F.13 compares the impacts of small and large minimum wage increases.

Table F.8: Heterogeneity in Firm Outcomes by Firm Size

	Q1	Baseline Revenue Quartiles		
		Q2	Q3	Q4
Revenues	0.0296 (0.0234)	0.0316* (0.0173)	0.0212 (0.0152)	0.0320** (0.0137)
Wage bill	0.0010*** (0.0035)	0.0188*** (0.0031)	0.0149*** (0.0028)	0.0128*** (0.0035)
Profits	0.0005 (0.0066)	0.0027 (0.0032)	-0.0004 (0.0028)	0.0007 (0.0022)
Employment (levels)	-0.222 (0.156)	-0.348 (0.224)	-1.966*** (0.611)	-3.952** (1.801)
Own Wage Elasticity	-0.085 (0.134)	-0.012 (0.167)	-0.417*** (0.148)	-0.366** (0.162)
Firm exit	0.0123 (0.0079)	0.0043 (0.0170)	-0.0079 (0.0123)	-0.0040 (0.0069)
N	123,784	135,370	135,600	135,656

Note: This table presents estimated effects of the minimum wage increases on independent businesses by baseline revenue quartiles. Firms are categorized by their position in the pre-reform (year $s-1$) revenue distribution; the average revenues by quartile are Q1=\$198,240, Q2=\$543,746, Q3=\$1,156,525, Q4=\$8,741,782. In the first three rows the outcome is the variable listed in each row scaled by baseline ($s-1$) revenue. In the next two rows the outcomes are scaled by the firm's baseline value of the listed variable. The employment outcome is the number of distinct employment relationships at the firm. The own-wage elasticity (OWE) is calculated by dividing the estimated percent change in employment by the percent change in average wages, with standard errors calculated using the Delta method. "Firm exit" is a binary outcome equal to one if a firm filed a tax return in year $s-1$ but did not in year $s+4$; as such negative coefficients correspond with increased exit. Each estimate is from a firm-level regression of Equation (1) with a linear probability model used to estimate exit effects; the reported coefficients are estimated effects on the outcomes four years after the minimum wage increase, $s+4$, relative to the base year, $s-1$. Regressions where outcomes are scaled by baseline revenue are weighted by log baseline revenue; when estimating percent changes for the OWE, regressions are scaled by the baseline value of the outcome. In addition to firm-cohort and cohort-by-year fixed effects, the specifications include categories of baseline firm size (# of workers), deciles of baseline firm value-added, baseline two-digit industry, quintiles of county density, and quintiles of county employment rates – all flexibly interacted with year to allow for differential time trends. OLS outcomes are winsorized from above and below at the 1% level. Standard errors are clustered at the state-by-cohort level.

Table F.9: Heterogeneity in Firm Outcomes - Low-Earning Labor Share of Variable Costs

	Quartiles: Low wage wage bill / variable costs			
	Q1	Q2	Q3	Q4
<u>Outcome as share of baseline revenue</u>				
Revenues	0.0394** (0.0194)	0.0252* (0.0131)	0.0315** (0.0136)	0.0365** (0.0184)
Wage bill	0.0049 (0.0030)	0.0111*** (0.0034)	0.0189*** (0.0038)	0.0208*** (0.0041)
Profits	0.0010 (0.0035)	0.0002 (0.0021)	0.0003 (0.0031)	-0.0004 (0.0047)
<u>Percent changes</u>				
Wage bill	0.0468*** (0.0082)	0.0652*** (0.0039)	0.0848*** (0.0041)	0.1190*** (0.0044)
Employment	-0.0146* (0.0082)	-0.0134 (0.0102)	-0.0220 (0.0144)	-0.0424*** (0.0146)
OWE	-0.312* (0.184)	-0.206 (0.157)	-0.259 (0.170)	-0.356** (0.123)
<u>Other</u>				
Employment (levels)	-1.219** (0.595)	-1.419** (0.647)	-0.813 (0.943)	-2.716*** (0.795)
Firm exit	0.0096** (0.0048)	-0.0027 (0.0062)	0.0034 (0.0104)	0.0040 (0.0123)
N	122,754	129,906	129,960	129,763

Note: This table presents estimated effects of the minimum wage increases on independent businesses by baseline reliance on low-earning labor. The low-earning labor share of variable costs is the ratio of the firm's low-earning wage bill to its total variable costs. The low-earning wage bill is the total wages paid by each firm to employees that earn $< \$20,000$ across all jobs in the base year $s-1$ and $< \$25,000$ (including not working) in year $s-2$. Variable costs are defined as $COGS + wagebill$. In the first three rows the outcome is the variable listed in each row scaled by baseline $(s-1)$ revenue. In the next two rows the outcomes are scaled by the firm's baseline value of the listed variable. The own-wage elasticity (OWE) is calculated by dividing the estimated percent change in employment by the percent change in average wages, with standard errors calculated using the Delta method. The employment outcome is the number of distinct employment relationships at the firm. "Firm exit" is a binary outcome equal to one if a firm filed a tax return in year $s-1$ but did not in year $s+4$; as such negative coefficients correspond with increased exit. Each estimate is from a firm-level regression of Equation (1) with a linear probability model used to estimate exit effects; the reported coefficients are estimated effects on the outcomes four years after the minimum wage increase, $s+4$, relative to the base year, $s-1$. Regressions where outcomes are scaled by baseline revenue are weighted by log baseline revenue; when estimating percent changes regressions are scaled by the baseline value of the outcome. In addition to firm-cohort and cohort-by-year fixed effects, the specifications include categories of baseline firm size (# of workers), deciles of baseline firm value-added, baseline two-digit industry, quintiles of county density, and quintiles of county employment rates – all flexibly interacted with year to allow for differential time trends. OLS outcomes are winsorized from above and below at the 1% level. Standard errors are clustered at the state-by-cohort level.

Table F.10: Heterogeneity in Firm Outcomes by Baseline Productivity

	Baseline Value-added per worker quartiles			
	Q1	Q2	Q3	Q4
Revenues	0.0229 (0.0169)	0.0153 (0.0125)	0.0280** (0.0132)	0.0421* (0.0212)
Wage bill	0.0187*** (0.0038)	0.0176*** (0.0034)	0.0154*** (0.0032)	0.0085*** (0.0025)
Profits	0.0014 (0.0029)	-0.0013 (0.0022)	-0.0014 (0.0028)	0.0016 (0.0054)
Employment	-2.846* (1.512)	-3.791*** (1.004)	-0.850 (0.588)	-0.502** (0.228)
Firm exit	-0.0021 (0.0109)	-0.0179 (0.0127)	-0.0088 (0.0107)	-0.00004 (0.0092)
Own wage elasticity	-0.395** (0.164)	-0.417** (0.164)	-0.259 (0.162)	0.092 (0.262)
N	115,304	116,782	116,802	116,816

Note: This table presents estimated effects of the minimum wage increases on independent businesses by baseline quartile of value-added per worker. Value-added per worker is revenue less COGS scaled by the number of employees. In the first three rows the outcome is the variable listed in each row scaled by baseline ($s - 1$) revenue. In the next two rows the outcomes are scaled by the firm's baseline value of the listed variable. The employment outcome is the number of distinct employment relationships at the firm. The own-wage elasticity (OWE) is calculated by dividing the estimated percent change in employment by the percent change in average wages, with standard errors calculated using the Delta method. "Firm exit" is a binary outcome equal to one if a firm filed a tax return in year $s - 1$ but did not in year $s + 4$; as such negative coefficients correspond with increased exit. Each estimate is from a firm-level regression of Equation (1) with a linear probability model used to estimate exit effects; the reported coefficients are estimated effects on the outcomes four years after the minimum wage increase, $s + 4$, relative to the base year, $s - 1$. Regressions where outcomes are scaled by baseline revenue are weighted by log baseline revenue; when estimating percent changes for the OWE, regressions are scaled by the baseline value of the outcome. In addition to firm-cohort and cohort-by-year fixed effects, the specifications include categories of baseline firm size (# of workers), deciles of baseline firm value-added, baseline two-digit industry, quintiles of county density, and quintiles of county employment rates – all flexibly interacted with year to allow for differential time trends. OLS outcomes are winsorized from above and below at the 1% level. Standard errors are clustered at the state-by-cohort level.

Table F.11: Robustness Estimated Effects of Minimum Wage by Event Years and Subsets of Treatment States

	Event year (of Min Wage change)				All event years,
	2013	2014	2015	2016 [†]	excluding CA
<u>Outcome as share of baseline revenue</u>					
Revenues	0.0145 (0.0139)	0.0488* (0.0282)	0.0197 (0.0147)	0.0637*** (0.0102)	0.0143* (0.0076)
Wage bill	0.0153*** (0.0022)	0.0164*** (0.0056)	0.0114*** (0.0041)	0.0258*** (0.0023)	0.0105*** (0.0027)
Intermediate Costs	0.0082 (0.0083)	0.0210 (0.0138)	0.096 (0.0085)	0.0297*** (0.0057)	0.0075 (0.0046)
Profits	-0.0080** (0.0024)	0.0060 (0.0038)	-0.0017 (0.0030)	0.0060** (0.0026)	-0.0027 (0.0020)
<u>Employment</u>					
Levels	-3.688*** (0.540)	-0.984 (0.919)	-0.822 (0.631)	0.242 (0.398)	-1.417* (0.757)
Logs	-0.0406** (0.0175)	-0.0261 (0.0229)	-0.0276 (0.0182)	0.0008 (0.0115)	-0.0172 (0.0124)
Own wage elasticity	-0.406** (0.177)	-0.348 (0.329)	-0.298 (0.201)	0.005 (0.072)	-0.265 (0.195)
N	162,154	202,786	165,472	135,880	487,791

Note: This table presents estimated effects of the minimum wage increases on independent businesses for subsets of the treatment group. The first four columns correspond to the estimated effects on firms in states that initially increased their minimum wage in the given year, as depicted in Figure 1 and detailed in Appendix Table A.1. The last column shows the estimated effects when excluding firms in California, the largest state in the sample ($\approx 24\%$ of firms in the treatment group). Employment effects are estimated in levels, natural logs and as own-wage elasticities (OWEs), where the OWE is estimated by dividing the estimated change in log employment by the estimated change in log average earnings, with standard errors calculated using the Delta method. Each estimate is from a firm-level regression of Equation (1); the coefficients are estimated effects on the outcomes four years after the minimum wage increase, $s + 4$, relative to the base year, $s - 1$. Regressions where outcomes are scaled by baseline revenue are weighted by log baseline revenue; when outcomes are in logs, regressions are scaled by the baseline value of the outcome. In addition to firm-cohort and cohort-by-year fixed effects, the specifications include categories of baseline firm size (# of workers), deciles of baseline firm value-added, baseline two-digit industry, quintiles of county density, and quintiles of county employment rates – all flexibly interacted with with year to allow for differential time trends. All outcomes are winsorized from above and below at the 1% level. Standard errors are clustered at the state-by-cohort level.

[†]The coefficients for the 2016 events are estimated for year $t+3$ as year $t+4$ would be 2020 which is outside of our sample period. The only treatment state for the 2016 event year is Maine.

Table F.12: Robustness - Estimated Effects of Minimum Wage by Event

	Scaled by Baseline Revenue				Percent Changes			Employment		
	Revenue	Wage bill	Material costs	Profits	Wage bill	Material costs	valadd/worker	Level	Percent	N
NY	0.0145 (0.0134)	0.0153*** (0.0021)	0.0084 (0.0081)	-0.0081*** (0.0023)	0.0362*** (0.0119)	0.0319* (0.0158)	0.0022 (0.0063)	-3.827*** (0.495)	-.0414*** (0.0121)	160,032
RI	0.0349* (0.0182)	0.0165*** (0.0039)	0.0185* (0.0105)	-0.0056 (0.0040)	0.0365 (0.0254)	0.0129 (0.0247)	0.0268*** (0.0080)	-1.853* (0.996)	-0.0003 (0.0186)	136,878
CA	0.0897*** (0.0126)	0.0259*** (0.0020)	0.0439*** (0.0073)	0.0122*** (0.0024)	0.0990*** (0.0118)	0.0740*** (0.0128)	0.0881*** (0.0078)	-1.875*** (0.558)	-0.0332*** (0.0110)	166,668
CT	-0.0258* (0.0136)	0.0036 (0.0024)	-0.0214** (0.0087)	-0.0052** (0.0024)	0.0276 (0.0182)	-0.0087 (0.0174)	-0.0141 (0.0091)	-0.163 (0.711)	-0.0147 (0.0165)	138,296
DC	-0.0541*** (0.0133)	-0.0153*** (0.0025)	-0.0214** (0.0087)	-0.0006 (0.0023)	-0.0920*** (0.0196)	-0.0902*** (0.0181)	0.0306*** (0.0105)	-7.349*** (0.922)	-0.167*** (0.0172)	135,342
DE	-0.0090 (0.0124)	0.0019 (0.0019)	-0.0159** (0.0070)	-0.0093** (0.0038)	-0.0198* (0.0097)	-0.0010 (0.0115)	0.0013 (0.0086)	1.582*** (0.499)	0.0256* (0.0132)	135,646
MI	0.0137 (0.0113)	0.0148*** (0.0019)	0.0042 (0.0071)	0.0007 (0.0019)	0.0951*** (0.0118)	0.0347** (0.0131)	0.0400*** (0.0057)	2.473*** (0.476)	0.0612*** (0.0113)	146,648
MN	0.0044 (0.0124)	0.0137*** (0.0023)	-0.0010 (0.0073)	-0.0001 (0.0027)	0.0942*** (0.0151)	0.0289* (0.0155)	0.0094 (0.0062)	-0.806 (0.641)	0.0033 (0.0149)	140,996
NJ	-0.0080 (0.0147)	-0.0060** (0.0025)	-0.0008 (0.0095)	0.0039 (0.0025)	-0.0259 (0.0190)	0.0017 (0.0190)	0.0021 (0.0093)	-2.609*** (0.682)	-0.0254 (0.0164)	146,226
AK	0.0620** (0.0234)	0.0178*** (0.0050)	0.0185 (0.0121)	0.0184*** (0.0055)	-0.0177 (0.0367)	-0.0172 (0.0280)	0.0158 (0.0172)	1.569 (1.097)	-0.0082 (0.0270)	135,198
AR	-0.0001 (0.0139)	0.0029 (0.0027)	-0.0008 (0.0074)	0.0040 (0.0030)	0.0293* (0.0163)	0.0150 (0.0123)	-0.0076 (0.0092)	-0.301 (0.547)	-0.0121 (0.0127)	137,078
CHI	0.0374** (0.0178)	0.0130*** (0.0028)	0.0222* (0.0117)	0.004 (0.0037)	0.0580** (0.0202)	-0.0174 (0.0154)	0.0442*** (0.0662)	-1.951*** (0.569)	-0.0806*** (0.0163)	137,908
HI	0.0432*** (0.0119)	0.0130*** (0.0023)	0.0146** (0.0068)	0.0041 (0.0027)	0.0568*** (0.0160)	-0.0059 (0.0112)	0.0278** (0.0100)	-3.205*** (0.465)	-0.101*** (0.0130)	135,786
MA	0.0357** (0.0149)	0.0215*** (0.0024)	0.0176* (0.0097)	-0.0024 (0.0031)	0.0837*** (0.0181)	0.0293** (0.0138)	0.0414*** (0.0119)	-0.026 (0.510)	-0.0167 (0.0136)	144,052
MD	0.0107 (0.0152)	0.0061** (0.0024)	0.0079 (0.0095)	-0.0024 (0.0032)	0.0445** (0.0183)	0.0377** (0.0139)	0.0131 (0.0116)	-0.743 (0.541)	-0.0417*** (0.0137)	140,046
NE	-0.0065 (0.0128)	0.0053* (0.0027)	-0.0041 (0.0068)	-0.0104*** (0.0026)	0.0615*** (0.0163)	0.0206 (0.0139)	-0.0300*** (0.0085)	-0.368 (0.552)	-0.0013 (0.0126)	136,500
SD	0.0151 (0.0179)	0.0058 (0.0039)	0.0074 (0.0089)	-0.0024 (0.0042)	0.0190 (0.0255)	0.0009 (0.0180)	0.0372*** (0.0124)	-1.714* (0.941)	-0.0308 (0.0193)	135,768
WV	-0.0608*** (0.0125)	-0.0104*** (0.0024)	-0.0327*** (0.0066)	-0.0138*** (0.0029)	-0.0371** (0.0130)	-0.215** (0.0099)	-0.0396*** (0.0089)	-3.330*** (0.482)	-0.101*** (0.0106)	135,472
ME†	0.0637*** (0.0102)	0.0026*** (0.0023)	0.0297*** (0.0057)	0.0060** (0.0026)	0.124*** (0.0111)	0.0510*** (0.0100)	0.0681*** (0.0089)	0.242 (0.389)	-0.0004 (0.0115)	135,880

Note: This table presents estimated effects of the minimum wage increases on independent businesses separately for each event, as depicted in Figure 1 and detailed in Appendix Table A.1. Each estimate is from a firm-level regression of Equation (1); the coefficients are estimated effects on the outcomes four years after the minimum wage increase, $s + 4$, relative to the base year, $s - 1$. Regressions where outcomes are scaled by baseline revenue are weighted by log baseline revenue; when estimates are percent changes regressions are scaled by the baseline value of the outcome. Employment effects are estimated in levels, natural logs. In addition to firm-cohort and cohort-by-year fixed effects, the specifications include categories of baseline firm size (# of workers), deciles of baseline firm value-added, baseline two-digit industry, quintiles of county density, and quintiles of county employment rates – all flexibly interacted with with year to allow for differential time trends. All outcomes are winsorized from above and below at the 1% level. Standard errors are clustered at the state-by-cohort level. †The coefficients for the 2016 event, Maine (ME) are estimated for year $t+3$ as year $t+4$ would be 2020 which is outside of our sample period.

Table F.13: Estimated effects, Large v. Small Minimum Wage Increases

	Small	Big
<u>Outcome as share of baseline revenue</u>		
Revenues	0.0055 (0.0104)	0.0447** (0.0201)
Wage bill	0.0056 (0.0042)	0.0186*** (0.0028)
Intermediate Costs	0.0031 (0.0059)	0.0198** (0.0096)
Profits	0.0001 (0.0020)	0.0010 (0.0041)
<u>Employment</u>		
Levels	-0.422 (1.098)	-2.042*** (0.593)
Logs	-0.0077 (0.0241)	-0.0305*** (0.0085)
Own wage elasticity	-0.204 (0.641)	-0.293** (0.088)
<u>Percent changes</u>		
Wage bill	0.0336 (0.0208)	0.103*** (0.0128)
Intermediate Costs	-0.0017 (0.0092)	0.0343* (0.0205)
Wages	0.0377*** (0.0084)	0.104*** (0.0114)
N	444,368	589,348

Note: This table examines the impact of minimum wage increases on independent businesses by size of the minimum wage increase. Small minimum wage increases raise the minimum wage by less than 30% over four years, whereas large increases raise it by 30% or more over four years. Employment effects are estimated in levels, natural logs and as own-wage elasticities (OWEs), where the OWE is estimated by dividing the estimated change in log employment by the estimated change in log average earnings, with standard errors calculated using the Delta method. Each estimate is from a firm-level regression of Equation (1); the coefficients are estimated effects on the outcomes four years after the minimum wage increase, $s + 4$, relative to the base year, $s - 1$. Regressions where outcomes are scaled by baseline revenue are weighted by log baseline revenue; when outcomes are in logs, regressions are scaled by the baseline value of the outcome. In addition to firm-cohort and cohort-by-year fixed effects, the specifications include categories of baseline firm size (# of workers), deciles of baseline firm value-added, baseline two-digit industry, quintiles of county density, and quintiles of county employment rates – all flexibly interacted with year to allow for differential time trends. All outcomes are winsorized from above and below at the 1% level. Standard errors are clustered at the state-by-cohort level.

G. Firm Analysis by Industry

The firm impacts presented in the main results largely average over all independent businesses, though Table 2 reports incidence analysis separately for restaurants and other highly exposed firms. This appendix presents more results broken down by industry. Appendix Table G.14 reports summary statistics for restaurants and other highly exposed independent firms while Appendix Figures G.2 and G.3 trace the impacts of the minimum wage increases on revenues, wage bills and profits for restaurants and other highly exposed independent firms.

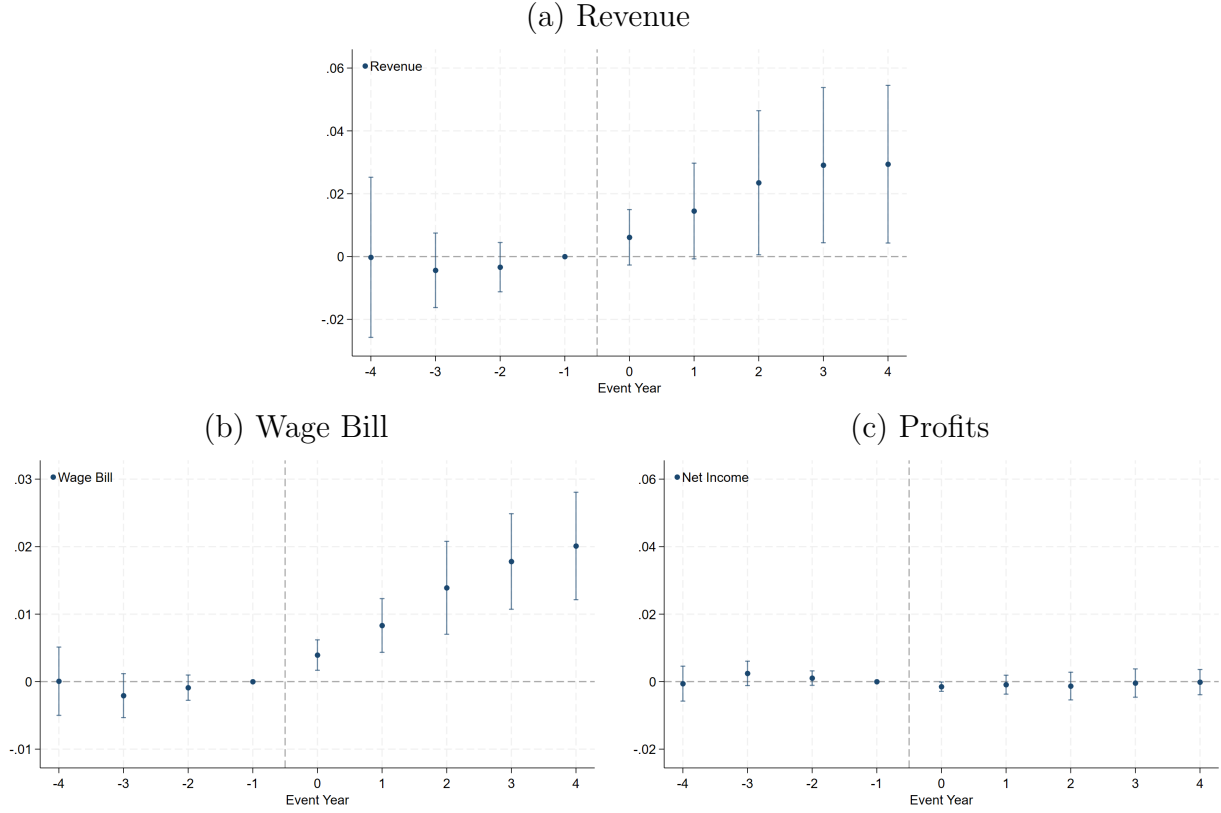
Appendix Table G.15 further breaks down the “other highly exposed independent firms” category, reporting outcomes by four-digit NAICS industry. The corresponding summary statistics by four-digit NAICS industry are reported in Appendix Table G.16.

Table G.14: Summary Statistics - Industry Comparison

	Means (base year)	
	Restaurants	Other/retail
Revenue (\$)	1,662,136	1,973,653
Wage Bill (\$)	420,194	256,465
Owner Income (\$)	126,300	122,499
Employees	51	36
<u>Cost structure:</u>		
Share low earning workers	0.278	0.278
Wage bill / revenues	0.252	0.129
COGS / revenues	0.571	0.647
Variable costs / revenues	0.833	0.789
Low earning labor / variable costs	0.054	0.039

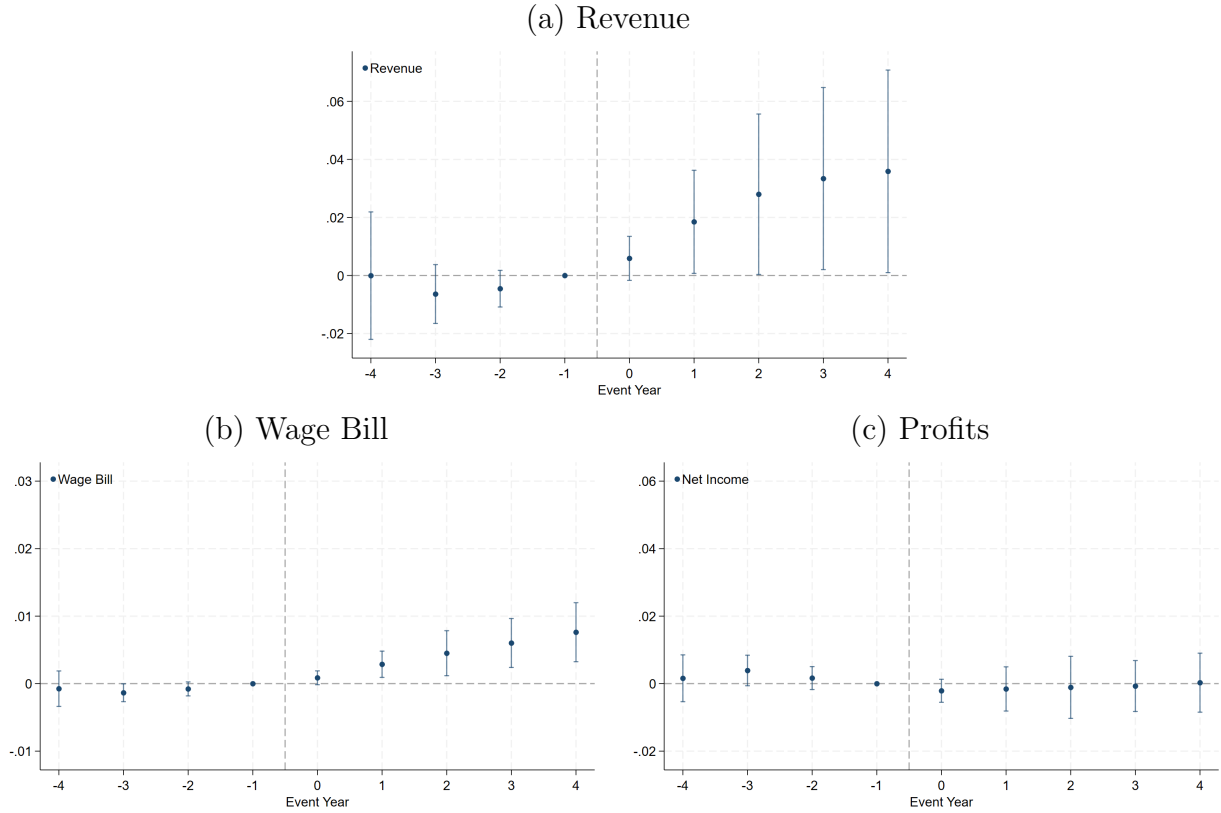
Note: This table shows pre-reform base year averages for independent businesses in highly exposed industries in treatment states, separately for restaurants and other, non-restaurant, highly exposed industries. Share of low earning workers is the share of employees at baseline that earn < \$20,000 across all jobs in the base year $s-1$ and < \$25,000 (including not working) in year $s-2$. Variable costs are defined as COGS+wage bill, and “low earning labor” is the wages paid to low-earning workers.

Figure G.2: Firm Outcomes Scaled by Baseline Revenues - Restaurants



Note: This figure plots the estimated effects of the minimum wage increases on independent restaurants, an industry highly exposed to the minimum wage. Plotted coefficients are estimates of β_s from (1) where the outcomes are firm revenues, wage bills and profits all scaled by revenue in base year $s - 1$. The coefficients trace the difference over time in scaled revenue relative to the base year $s - 1$ between firms in treated and untreated states. In addition to firm-cohort and cohort-by-year fixed effects, the specifications include categories of baseline firm size (# of workers), deciles of baseline firm value-added, baseline two-digit industry, quintiles of county density, and quintiles of county employment rates – all flexibly interacted with year to allow for differential time trends. Regressions are weighted by log base year revenue and outcomes are winsorized at the 99th percentile. For each point estimate the 95% confidence interval is marked with standard errors clustered at the state-by-cohort level.

Figure G.3: Firm Outcomes Scaled by Baseline Revenues - Other Exposed Industries



Note: This figure plots the estimated effects of the minimum wage increases on independent business in highly exposed industries, other than restaurants. Plotted coefficients are estimates of β_s from (1) where the outcomes are firm revenues, wage bills and profits all scaled by revenue in base year $s - 1$. The coefficients trace the difference over time in scaled revenue relative to the base year $s - 1$ between firms in treated and untreated states. In addition to firm-cohort and cohort-by-year fixed effects, the specifications include categories of baseline firm size (# of workers), deciles of baseline firm value-added, baseline two-digit industry, quintiles of county density, and quintiles of county employment rates – all flexibly interacted with year to allow for differential time trends. Regressions are weighted by log base year revenue and outcomes are winsorized at the 99th percentile. For each point estimate the 95% confidence interval is marked with standard errors clustered at the state-by-cohort level.

Table G.15: Estimated Effects Across Detailed Industry Groups

	All exposed	Restaurants	Retail	Arts, entertainment and recreation	Accommodations
<u>Outcomes scaled by baseline revenues</u>					
Revenue	0.0331** (0.0149)	0.0294** (0.0128)	0.0476*** (0.0156)	0.0101 (0.0186)	0.0367 (0.0249)
Wage bill	0.0143*** (0.0032)	0.0201*** (0.0041)	0.0057*** (0.0012)	0.0041 (0.0031)	0.0139*** (0.0038)
COGS	0.0131*** (0.0038)	0.0053 (0.0036)	0.0400*** (0.0105)	-0.0025 (0.0034)	0.0095*** (0.0028)
Owner Profits	0.0002 (0.0029)	-0.0001 (0.0019)	0.0014 (0.0026)	-0.00006 (0.0039)	0.0004 (0.0082)
<u>Percent changes</u>					
Wage bill	0.0599*** (0.0146)	0.0661*** (0.0158)	0.0576*** (0.0132)	0.0055 (0.0222)	0.0544*** (0.0194)
COGS	0.0373*** (0.0101)	0.0097 (0.0099)	0.0640*** (0.0130)	-0.0031 (0.0377)	0.0569** (0.0239)
<u>Levels</u>					
Employment	-1.496** (0.602)	-2.306** (0.915)	-0.332 (0.411)	-1.248** (0.505)	-0.501 (0.459)
N firms	134,974	68,193	26,709	15,327	23,817

Note: This table presents estimated effects of the minimum wage increases on independent businesses by major industry group within the highly exposed industries. Each column is for a 2-digit NAICS industry. In the first four rows the outcome is the variable listed in each row scaled by baseline ($s - 1$) revenue. In the next two rows the outcomes are scaled by the firm's baseline value of the listed variable. The employment outcome is the number of distinct employment relationships at the firm. Each estimate is from a firm-level regression of Equation (1); the reported coefficients are estimated effects on the outcomes four years after the minimum wage increase, $s + 4$, relative to the base year, $s - 1$. Regressions where outcomes are scaled by baseline revenue are weighted by log baseline revenue; when estimating percent changes regressions are scaled by the baseline value of the outcome. In addition to firm-cohort and cohort-by-year fixed effects, the specifications include categories of baseline firm size (# of workers), deciles of baseline firm value-added, baseline two-digit industry, quintiles of county density, and quintiles of county employment rates – all flexibly interacted with with year to allow for differential time trends. Standard errors are clustered at the state-by-cohort level. All raw reported firm values are winsorized from above and below at the 1% level. Regressions are weighted by log baseline revenue.

Table G.16: Detailed Industry Summary Statistics

	Means (base year)			
	Restaurants	Retail	Arts, entertainment and recreation	Accommodations
Revenue (\$)	1,662,136	2,638,705	1,389,296	1,609,919
Wage Bill (\$)	420,194	245,294	270,749	260,408
Owner Income (\$)	126,300	120,404	108,014	134,339
Employees	51	41	42	27
<u>Cost structure:</u>				
Share low earning workers	0.278	0.269	0.282	0.285
Wage bill / revenues	0.25	0.09	0.19	0.16
COGS / revenues	0.57	0.77	0.52	0.49
Variable costs / revenues	0.83	0.87	0.73	0.67
Low earning labor / variable costs	0.054	0.020	0.053	0.051
N	68,193	26,709	15,327	23,817

Note: This table shows pre-reform base year averages for independent businesses in treatment states, across the various major (2-digit NAICS) highly exposed industries included in the main firm analysis sample. Share of low earning workers is the share of employees at baseline that earn < \$20,000 across all jobs in the base year $s-1$ and < \$25,000 (including not working) in year $s-2$. Variable costs are defined as COGS + wage bill, and “low earning labor” is the wages paid to low-earning workers.

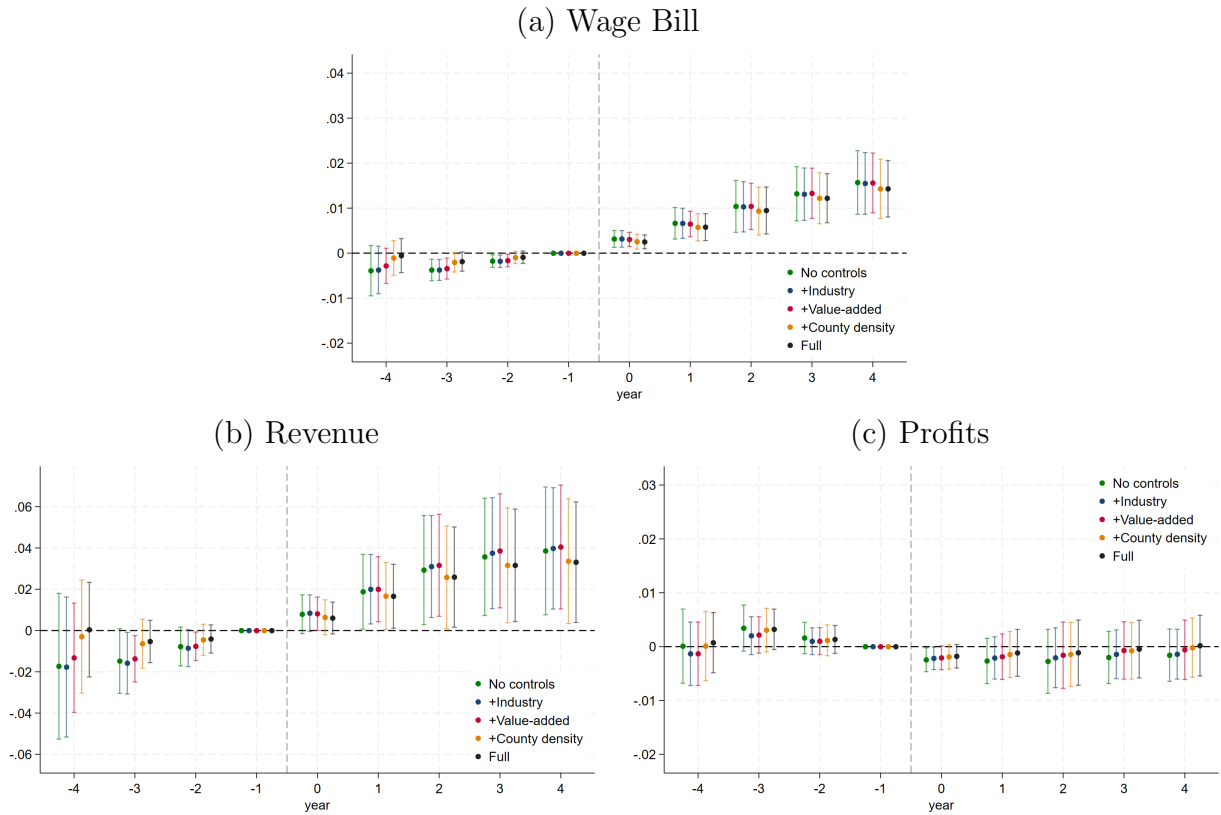
H. Robustness and Alternative Specifications: Firm-Level Results

The sensitivity of the main firm analysis results to alternative controls and measures is assessed below.

H.1. Controls

Appendix Figure H.4 presents pre-trend and impact estimates from regression models that do not include controls, control only for value-added deciles, then add county density controls as well, and finally include all controls.

Figure H.4: Robustness to Controls - Firm Outcomes Scaled by Baseline Revenues



Note: This figure plots the estimated effects the minimum wage increases on firm outcomes, using various sets of controls. Plotted coefficients are estimates of β_s from (1) where the outcomes are firm wage bills, revenues and profits, all scaled by revenue in base year $s-1$. The different series show estimates layering on control variables. The first estimates “no controls” include only firm-cohort and cohort-by-year fixed effects, “+ industry” adds indicators for two-digit industry interacted with year, “+ Value-added” adds deciles of baseline firm value-added interacted with year, “+ county density” adds quintiles of county density interacted with year, and “Full” includes all of the controls in the main specification (i.e. adds controls for categories of baseline firm size (# of workers) and quintiles of county employment rates each interacted with year). Regressions are weighted by log base year revenue and scaled revenue ratios are winsorized at the 99th percentile. For each point estimate the 95% confidence interval is marked with standard errors clustered at the state-by-cohort level.

H.2. Percent Changes

Appendix Table H.17 presents estimates of the impact of state minimum wage increases on key firm outcomes measured as a percentage change relative to baseline; for example, among

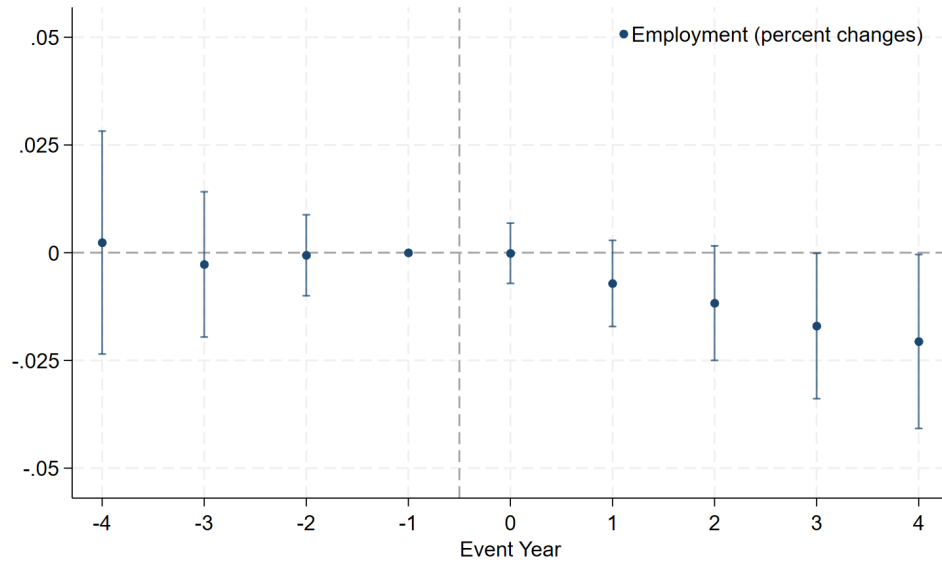
all firms in highly exposed industries, revenues increase by 3.59% following the analyzed state minimum wage increases. Appendix Figures H.6 and H.7 show the evolution of average firm wage bills and COGS as percentage changes relative to the base year. Estimates are provided for all pass-throughs in highly exposed industries as well as for restaurants and other exposed firms separately. All estimates are from the balanced panel.

Table H.17: Firm Outcomes - Percent Changes

	Percent Changes $s-1$ to $s+4$		
	All exposed	Restaurants	Other exposed/retail
Revenues	0.0331** (0.0149)	0.0294** (0.0128)	0.0359** (0.0178)
Wage bill	0.0599*** (0.0146)	0.0661*** (0.0158)	0.0439*** (0.0135)
COGS	0.0373*** (0.0101)	0.0097 (0.0099)	0.0531*** (0.0131)
Other deductions	0.0233 (0.0200)	0.0140 (0.0203)	0.0300 (0.0209)
Variable costs	0.0434*** (0.0132)	0.0327** (0.0143)	0.0492*** (0.0136)
Employment	-0.0206** (0.0103)	-0.0345*** (0.0119)	-0.0003 (0.0045)
Avg. wages	0.0811*** (0.0134)	0.095*** (0.0148)	0.0643*** (0.0119)
Own wage elasticity	-0.254* (0.134)	-0.364** (0.138)	-0.0004 (0.149)
N firms	134,974	68,193	66,781

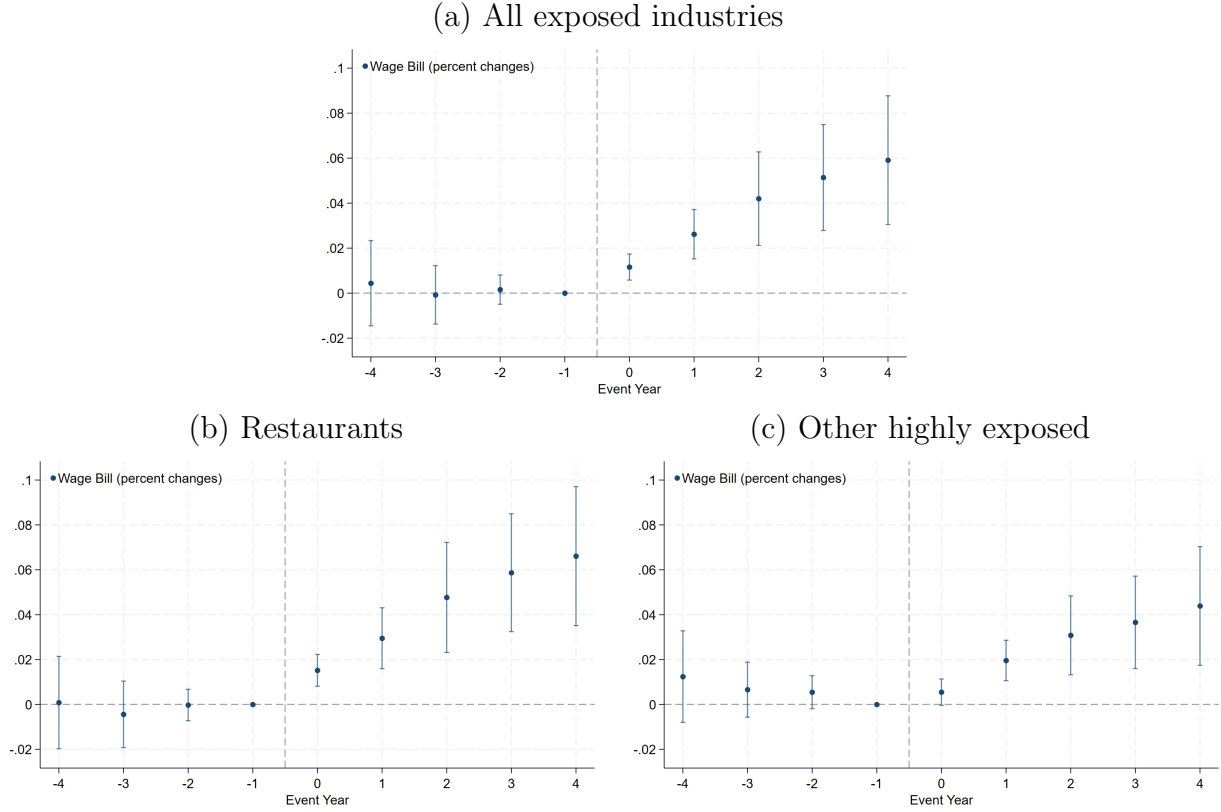
Note: This table presents estimated effects of the minimum wage increases on independent businesses in all highly exposed industries (main sample), then separately for restaurants and other, non-restaurant, highly exposed industries. Each estimate is from a firm-level regression of Equation (1); the coefficients are estimated effects on the outcomes four years after the minimum wage increase, $s + 4$, relative to the base year, $s - 1$. To estimate percent changes, the outcomes in year t are scaled by the baseline value of the outcome. Regressions are weighted by the baseline value of the outcome. The own-wage elasticity (OWE) is calculated by dividing the estimated percent change in employment by the percent change in average wages, with standard errors calculated using the Delta method. In addition to firm-cohort and cohort-by-year fixed effects, the specifications include categories of baseline firm size (# of workers), deciles of baseline firm value-added, baseline two-digit industry, quintiles of county density, and quintiles of county employment rates – all flexibly interacted with with year to allow for differential time trends. Standard errors are clustered at the state-by-cohort level.

Figure H.5: Effect of the Minimum Wage on Percent Changes in Employment



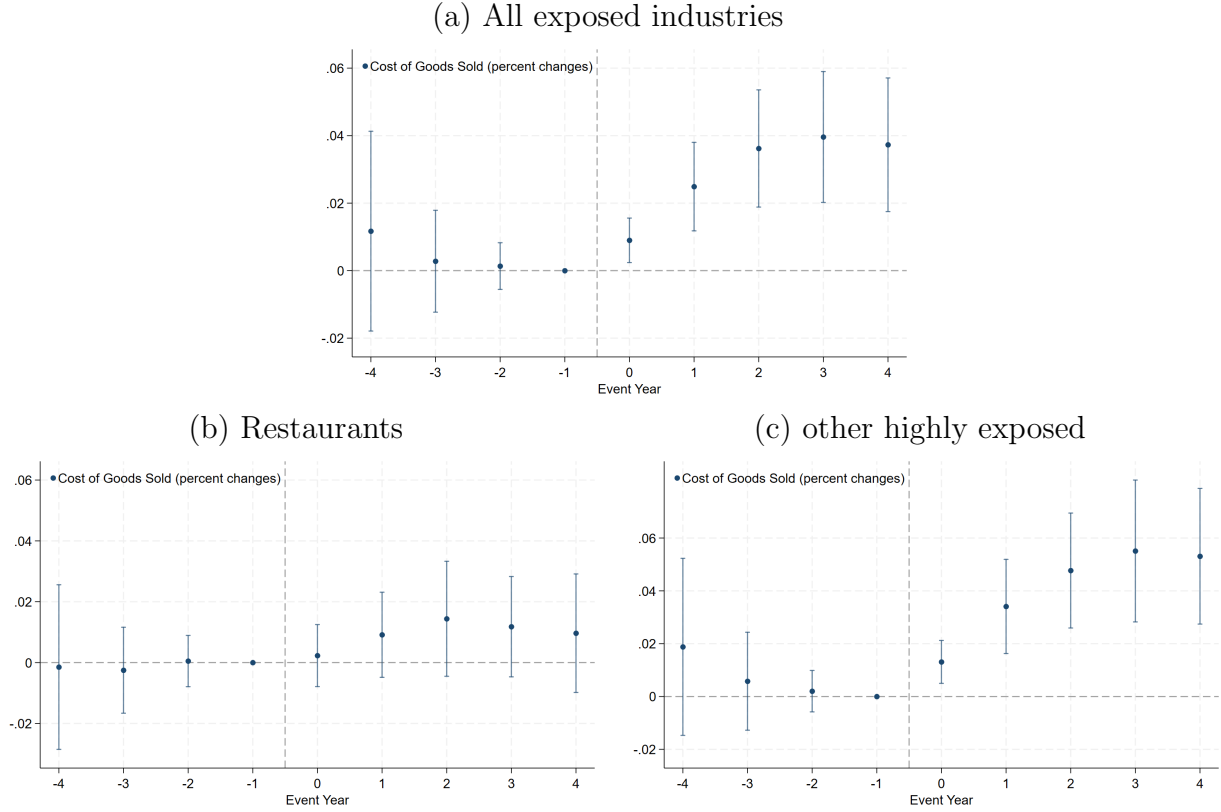
Note: This figure describes the percent change in employment relationships due to the minimum wage. Our employment measure counts the number of W-2s attached to a firm each year. Plotted coefficients are estimates of β_s from (1) where the outcome is log employment, so convey the estimated percent change employment relationships relative to base year $s-1$, for the average treated firm compared to the average control firm. In addition to firm-cohort and cohort-by-year fixed effects, the specifications include categories of baseline firm size (# of workers), deciles of baseline firm value-added, baseline two-digit industry, quintiles of county density, and quintiles of county employment rates – all flexibly interacted with year to allow for differential time trends. Regressions are weighted by base year firm employment. For each point estimate the 95% confidence interval is marked with standard errors clustered at the state-by-cohort level.

Figure H.6: Effect of the Minimum Wage on Percent Changes in Wage Bills



Note: This figure plots the estimated effects of the state minimum wage increases on firm wage bills. Plotted coefficients are estimates of β_s from (1) where the outcome is firm wage bill in year t scaled by base year wage bill. The coefficients trace the differential percent change in wage bills between firms in treated and untreated states, relative to the base year $s - 1$. In addition to firm-cohort and cohort-by-year fixed effects, the specifications include categories of baseline firm size (# of workers), deciles of baseline firm value-added, baseline two-digit industry, quintiles of county density, and quintiles of county employment rates – all flexibly interacted with year to allow for differential time trends. Regressions are weighted by base year wage bill and outcomes are winsorized at the 99th percentile. For each point estimate the 95% confidence interval is marked with standard errors clustered at the state-by-cohort level. Panel (a) presents the estimated effects for independent businesses in all highly exposed industries; Panel (b) for restaurants specifically; and Panel (c) for the other, non-restaurant, highly exposed industries.

Figure H.7: Effect of the Minimum Wage on Percent Changes in Cost of Goods Sold



Note: This figure plots the estimated effects of the state minimum wage increases on firm COGS. Plotted coefficients are estimates of β_s from (1) where the outcome is firm COGS in year t scaled by base year COGS. The coefficients trace the differential percent change in COGS between firms in treated and untreated states, relative to the base year $s - 1$. In addition to firm-cohort and cohort-by-year fixed effects, the specifications include categories of baseline firm size (# of workers), deciles of baseline firm value-added, baseline two-digit industry, quintiles of county density, and quintiles of county employment rates – all flexibly interacted with year to allow for differential time trends. Regressions are weighted by base year COGS and outcomes are winsorized at the 99th percentile. For each point estimate the 95% confidence interval is marked with standard errors clustered at the state-by-cohort level. Panel (a) presents the estimated effects for independent businesses in all highly exposed industries; Panel (b) for restaurants specifically; and Panel (c) for the other, non-restaurant, highly exposed industries.

H.3. Average productivity measured at the industry-level

Table H.18: Effects on Average Productivity and Cost Structure - Industry Level

	Wage bill/ revenue	Material costs/ revenue	Value added/ worker	Profits/ revenue	Employees
<u>Industry-level, market controls (main specification)</u>					
All firms	0.0075 (0.0176)	-0.0042 (0.0201)	0.0423 (0.100)	0.0046 (0.0068)	-1.838 (2.405)
Entrants	0.0133 (0.0126)	-0.0091 (0.0133)	0.132* (0.0680)	0.0130 (0.0106)	-3.970** (1.677)
Incumbents	0.0086 (0.0178)	-0.0028 (0.0209)	0.0316 (0.0985)	-0.0063 (0.0062)	-1.229 (2.563)
<u>Industry-level, no controls</u>					
All firms	0.0084 (0.0423)	-0.0040 (0.0498)	0.0413 (0.254)	0.0035 (0.0151)	-1.594 (5.984)
Entrants	0.0084 (0.0297)	-0.0086 (0.0318)	0.207 (0.171)	0.0081 (0.0219)	-3.985 (3.954)
Incumbents	0.0094 (0.0431)	-0.0035 (0.0516)	0.0414 (0.251)	-0.0035 (0.0150)	-1.421 (6.462)
<u>Firm-level, no controls</u>					
All firms	0.0072*** (0.0018)	-0.0037** (0.0017)	0.0475*** (0.0162)	0.0013 (0.0044)	-1.792*** (0.510)
Entrants	0.0049 (0.0037)	-0.0168** (0.0085)	0.249*** (0.0534)	0.0002 (0.0112)	-4.466*** (0.828)
Incumbents	0.0094*** (0.0018)	-0.0032 (0.0021)	0.0413** (0.0198)	-0.0040 (0.0027)	-0.421 (0.300)

Note: This table shows the statistics presented in Table 3, estimated in various ways. The top panel shows the estimates when the data is collapsed at the industry level, as opposed to the firm-level estimates presented in 3. Averages of each outcome variable are collapsed by industry, treatment state, cohort year and year. For value added per worker, the natural log of the outcome is collapsed. The regression includes the same controls as the main specification in 3 – two-digit industry and quintiles of county density, and quintiles of county employment rates flexibly interacted with with year. The point estimates are very similar to the firm-level estimates, while standard errors are much larger due to large reduction in sample size. The next panel reproduces the industry-level collapsed results, but without the market controls for density and employment rates. The third panel reproduces the firm-level estimates but without the market controls. Together, these panels show that the estimates are not substantially driven by the choice of control variables.

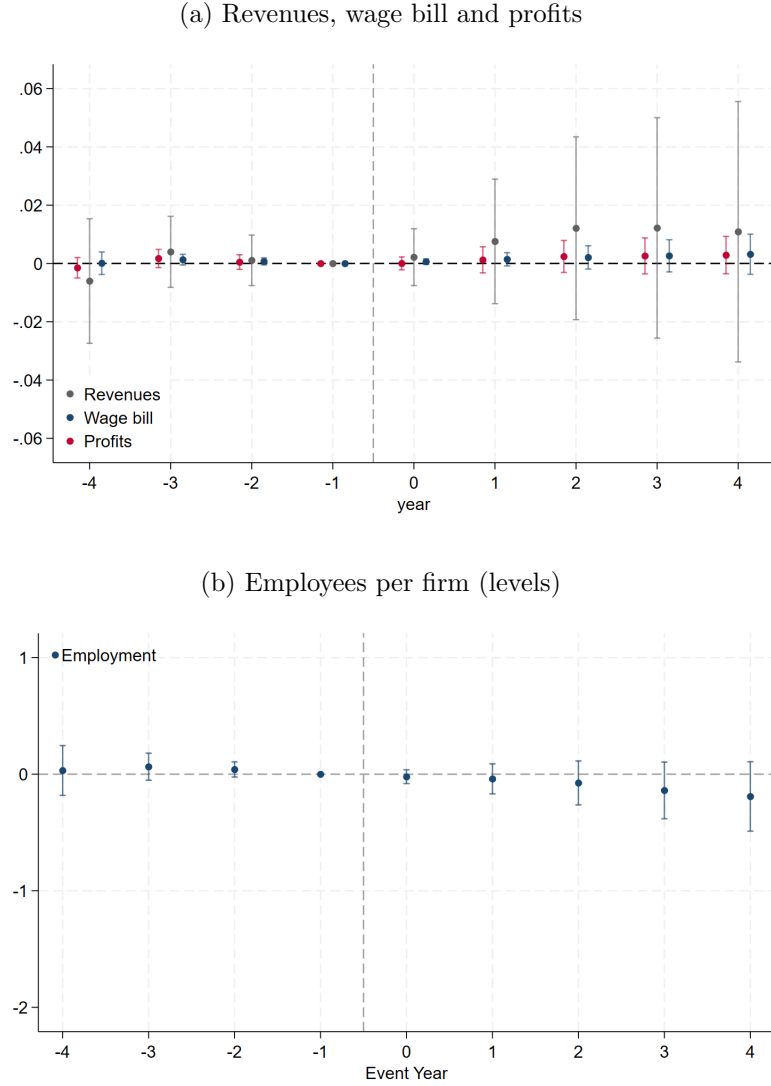
H.4. Placebo Tests

We may worry that the estimated impacts reported in Section 4 reflect not only the impact of the minimum wage but concomitant changes in the economic trajectories of treatment states relative to control states. Though the similarity of pre-trends between firms in states

that raised their minimum wages and states that left their wage floors unchanged bolsters our confidence in the comparability of firms in treatment and control states, we also conduct placebo tests to better understand the threat of concurrent macro events.

Appendix Figure [H.8](#) presents regression estimates of Equation (1) but for only firms in industries with low shares of minimum wage workers based on CPS data. Specifically, we include industries employing less than 1% of minimum wage workers. We conduct placebo tests for the impact of the minimum wage in these industries in treatment states (relative to control states) and find no statistically discernible impacts or point estimates of economically meaningful magnitudes for revenues, wage bills or profits. This concert of null estimates suggests that the estimated impacts in industries highly exposed to minimum wage policy are not attributable to macro events affecting the overall economies of states that raised their minimum wages.

Figure H.8: Placebo Test - Effect of Minimum Wage on Unexposed Industries



Note: This figure plots the estimated effects of the minimum wage increases on independent businesses operating in industries not highly exposed to the minimum wage. Plotted coefficients are estimates of β_s from Equation (1). In Panel A, the outcomes are firm revenues, wage bills and profits all scaled by revenue in base year $s - 1$, and regressions are weighted by log base year revenue. For Panel B, the outcome is the number of employees. The coefficients trace the difference over time in scaled revenue relative to the base year $s - 1$ between firms in treated and untreated states. In addition to firm-cohort and cohort-by-year fixed effects, the specifications include categories of baseline firm size (# of workers), deciles of baseline firm value-added, baseline two-digit industry, quintiles of county density, and quintiles of county employment rates – all flexibly interacted with year to allow for differential time trends. Outcomes are winsorized at the 99th percentile. For each point estimate the 95% confidence interval is marked with standard errors clustered at the state-by-cohort level.

Table H.19: Placebo Test - Effects on Average Productivity and Cost Structure among Unexposed Industries

	Wage bill/ revenue	Material costs/ revenue	Value added/ worker	Profits/ revenue	Employees
All firms	0.0008 (0.0009)	-0.0005 (0.0026)	0.0088 (0.0096)	-0.00002 (0.00321)	-0.078 (0.106)
percent change	0.6%	-0.2%	1.2%	1.3%	-0.5%
Entrants	0.0020 (0.00324)	-0.0102 (0.00954)	0.0129 (0.0470)	0.0052 (0.0106)	0.058 (0.177)
percent change	1.4%	-1.6%	1.3%	3.0%	0.4%
Incumbents	-0.0005 (0.0011)	0.0022 (0.0018)	0.0079 (0.0100)	-0.0039* (0.0022)	-0.208 (0.148)
percent change	-0.1%	0.3%	0.9%	1.9%	-0.2%

Note: This table shows estimated average changes in various metrics, estimated separately for all independent businesses in industries *not* highly exposed to the minimum wage increases ($\approx 2,257,602$ firms/year), Entrants (firms that are active four years after the minimum wage increases (year $s+4$) but did not exist prior (year $s-1$), and Incumbents (firms that were active in base year $s-1$ and remain active in year $s+4$, $\approx 1,385,973$ firms/year). Each estimate is from a firm-level regression using an unbalanced panel of all active independent firms in highly exposed industries using Equation (1). The coefficients are estimated effects on the outcomes four years after the minimum wage increase, $s+4$, relative to the base year, $s-1$. Each column represents an outcome variable. For value added per worker we use the natural log of the outcome to reduce noise and facilitate interpretation. In addition to firm-cohort and cohort-by-year fixed effects, the specifications include two-digit industry, quintiles of county density, and quintiles of county employment rates – all flexibly interacted with with year to allow for differential time trends. Standard errors are clustered at the state-by-cohort level. Outcomes are winsorized from above and below at the 1% level. To facilitate interpretation we include implied percent changes. For All firms and Incumbents, these are the coefficients evaluated at baseline averages; for Entrants coefficients are evaluated at the control mean in year $s+4$.

Table H.20: Placebo Test - Aggregate Effects on Independent Businesses in Unexposed Industries

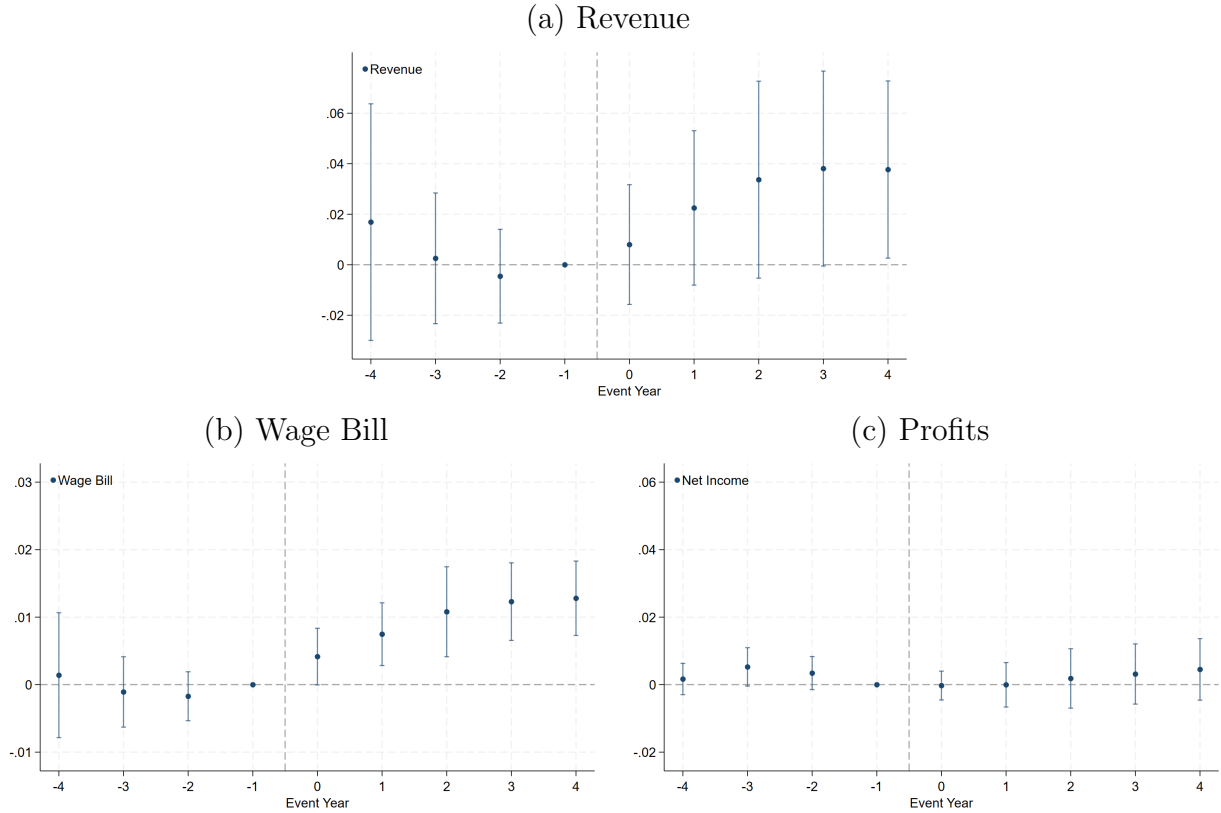
	Revenue	Wage bill	Material costs	Profits	Value added/worker
Aggregates	-0.0008	0.0010	-0.0066	0.0014	-0.0229
(share of baseline revenue)	(0.0227)	(0.0038)	(0.0146)	(0.0021)	(0.0915)

Note: This table presents estimates from the full, unbalanced panel of independent businesses in industries *not* highly exposed to the minimum wage increases ($\approx 2,257,602$ firms/year). Materials costs are the sum of COGS and Other Deductions. Value-added per worker is revenue less COGS scaled by the number of employees. The first row presents aggregate statistics summing over all independent firms in highly exposed industries. These estimates use collapsed state-by-industry data with state-by-industry totals of income statement items scaled by base year ($s - 1$) total revenue as the outcome variables. Section L details the specification. All aggregate regressions are weighted by baseline total revenue. All standard errors are clustered at the state-by-cohort level.

H.5. Unbalanced Panel of Independent Businesses

The main firm results presented in Tables 1 and 2 and Figures 2 through 5 use a balanced panel of independent businesses. Appendix Figure H.9 uses an unbalanced panel of firms to estimate the impacts of the minimum wage increases on firm revenues, wage bills and owner profits. The results are statistically indistinguishable from the main results using the balanced panel presented in the text. Estimates of the impact on firm outcomes in $s + 4$ are reported in Appendix Table E.7.

Figure H.9: Robustness - Unbalanced Panel, Firm Outcomes Scaled by Baseline Revenues



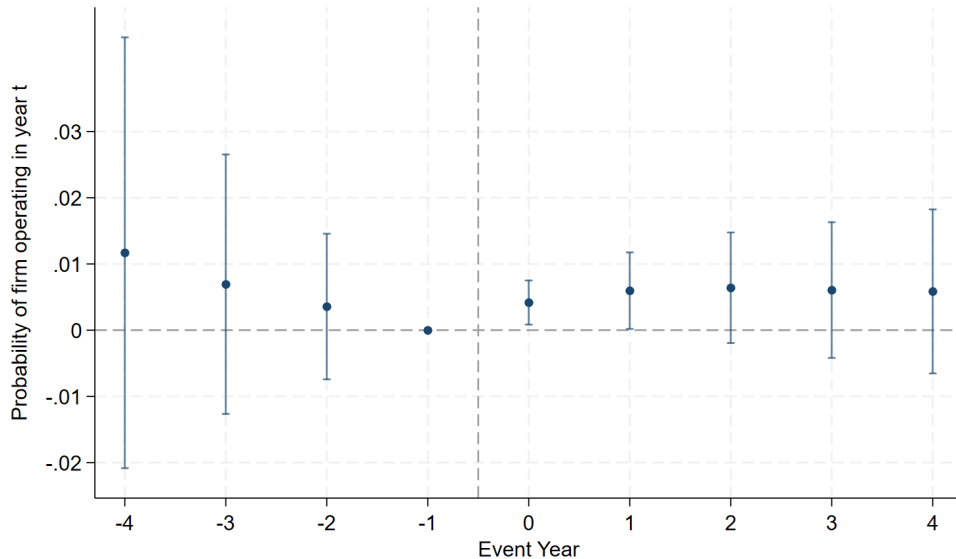
Note: This figure plots the estimated effects of the minimum wage increases on independent business in highly exposed industries, using an unbalanced panel. The unbalanced panel consists of firms that were active in pre-reform year $s-1$, but may have exited before year $s+4$ or entered after year $s-4$. Plotted coefficients are estimates of β_s from (1) where the outcomes are firm revenues, wage bills and profits all scaled by revenue in base year $s - 1$. The coefficients trace the difference over time in scaled revenue relative to the base year $s - 1$ between firms in treated and untreated states. In addition to firm-cohort and cohort-by-year fixed effects, the specifications include categories of baseline firm size (# of workers), deciles of baseline firm value-added, baseline two-digit industry, quintiles of county density, and quintiles of county employment rates – all flexibly interacted with year to allow for differential time trends. Regressions are weighted by log base year revenue and outcomes are winsorized at the 99th percentile. For each point estimate the 95% confidence interval is marked with standard errors clustered at the state-by-cohort level.

H.6. Alternative Firm Exit Specification

The entry and exit results plotted in Figure 6 are from regressions of count data. The construction of these data are detailed in Section L.5. Below, we offer estimates from an

alternative specification. The estimates plotted in Appendix Figure H.10 are the coefficients from linear probability model regression of Equation (1) where the outcome is a binary indicator equal to one if a firm that filed a tax return in time $s - 1$ does not file a return in time s .

Figure H.10: Effects on Firm Exit - Estimates from Firm-Level Regressions



Note: The figure above plots the estimated effects of the minimum wage increases on the exit rates of independent businesses operating in highly exposed industries. Plotted coefficients are estimates of β_s from linear probability model regressions of Equation (1) where the dependent variable is a binary indicator equal to one when a firm that filed a tax return in year $s - 1$ does not file a return in year s . In addition to firm-cohort and cohort-by-year fixed effects, the specifications include categories of baseline firm size (# of workers), deciles of baseline firm value-added, baseline two-digit industry, quintiles of county density, and quintiles of county employment rates – all flexibly interacted with year to allow for differential time trends. Regressions are weighted by log base year revenue and outcomes are winsorized at the 99th percentile. For each point estimate the 95% confidence interval is marked with standard errors clustered at the state-by-cohort level.

H.7. Excluding Firms with High Shares of Workers in Multiple States

Some independent businesses in our linked panel employ workers in more than one state. These may be enterprises that operate near a state border or they may be multi-establishment enterprises that operate in multiple states. In the latter case, we may worry that some workers may be employed in non-treated states and thus not subject to the wage floor increases, leading to downward biased estimates of the minimum wage's impact.

Appendix Table H.21 reports estimates of the four-year impact of the minimum wage increases on independent businesses excluding firms that report high shares of workers in multiple states. The upper row drops the 7% of firms that have more than 10% of their workers in control states, while the second row examines the 78% of firms with more than 70% of their workers in the same state. All estimates closely align with the main results presented in the paper.

Table H.21: Robustness - Excluding firms with higher shares of workers in multiple states

	Outcomes scaled by baseline revenues					
	Revenues	Wage bill	COGS	Profits	Employees	OWE
Excluding firms w/ >10% workers in control states	0.0340** (0.0151)	0.0149*** (0.0032)	0.0134*** (0.0037)	0.0002 (0.0030)	-1.424** (0.638)	-0.244* (0.138)
Excluding firms w/ <70% workers in reporting state	0.0358** (0.0151)	0.0147*** (0.0029)	0.0156*** (0.0039)	0.0002 (0.0032)	-0.538 (0.540)	-0.010 (0.168)

Note: This table shows estimates of the effects of the minimum wage on firm outcomes for subsamples of firms, defined by the share of the firms' workers that live in a different state than the reported firm address. The first row excludes the 7% of treatment firms that have more than 10% of their workers in control states. The second row includes only firms with more than 70% of their workers in the same state (78% of firms). All outcomes are estimated differences between firms in treatment and control states four years after the minimum wage increases (year $s+4$), relative to base year $s-1$, estimated using Equation (1). The outcomes for revenues, wage bill, COGS and profits are scaled by base year firm revenues and weighted by log baseline revenue. The employee estimates are in levels. The own-wage elasticity (OWE) is estimated by dividing the the estimated effects on log employment by the estimated effects on log average wages and standard errors are calculated using the Delta method. In addition to firm-cohort and cohort-by-year fixed effects, the specifications include categories of baseline firm size (# of workers), deciles of baseline firm value-added, baseline two-digit industry, quintiles of county density, and quintiles of county employment rates – all flexibly interacted with year to allow for differential time trends. Standard errors are clustered at the state-by-cohort level. Regression N for row 1 is 504,977 and N for row 2 is 490,183.

I. Additional Individual Results

The tables and figures below further examine the individual-level results including the sensitivity of the main individual analysis results to alternative controls and measurement choices, and investigate heterogeneity in responses.

I.1. Controls

Appendix Table I.22 and Appendix Table I.23 present four-year estimates of the impact of minimum wages on individual earnings and employment with different sets of controls, and using a “regression adjusted” estimator (see Appendix J for details of the regression adjustment). Estimates are very similar across controls.

Table I.22: Robustness - Individual Panels: Worker Earnings Effects

	Across regression controls				Regression adjusted	N
	(1)	(2)	(3)	(4)		
All low-earning	2,350*** (407)	2,177*** (372)	2,163*** (347)	1,473*** (304)	1,271*** (330)	2,545,172
Low-earning exposed industries	2,018*** (321)	1,898*** (312)	2,005*** (307)	1,532*** (287)	1,387*** (351)	644,636
All young	3,117*** (609)	2,884*** (547)	2,894*** (539)	1,995*** (488)	1,747*** (609)	1,235,008
All teen	2,247*** (345)	2,161*** (322)	2,121*** (352)	1,845*** (307)	1,571*** (393)	414,834
<u>Controls:</u>						
Baseline industry		X	X	X	X	
Age and age ²			X	X	X	
local market characteristics				X	X	

Note: This table presents the estimated effects of the minimum wage increases on earnings for baseline low-earning workers and young individuals, using various sets of controls and an alternative estimation strategy. The first four columns show estimates of β_s from Equation (2). Reported coefficients are estimates of the average change in annual earnings for each group between the base year prior to the policy change $s - 1$ and four years hence $s + 4$. Col. (1) shows results including only controls for individual-cohort and cohort-by-year fixed effects, Col. (2) adds controls for baseline industry, Col (3) adds baseline age and age-squared, Col. (4) adds baseline quintiles of county density and quintiles of county employment rates. All controls are flexibly interacted with year. Standard errors clustered at the state-by-cohort level. The last column shows results using a “regression adjusted” (RA) estimator, described in detail in Appendix J. The RA estimation uses the full set of controls and standard errors estimated using clustered bootstrapping with resampling at the state-cohort level.

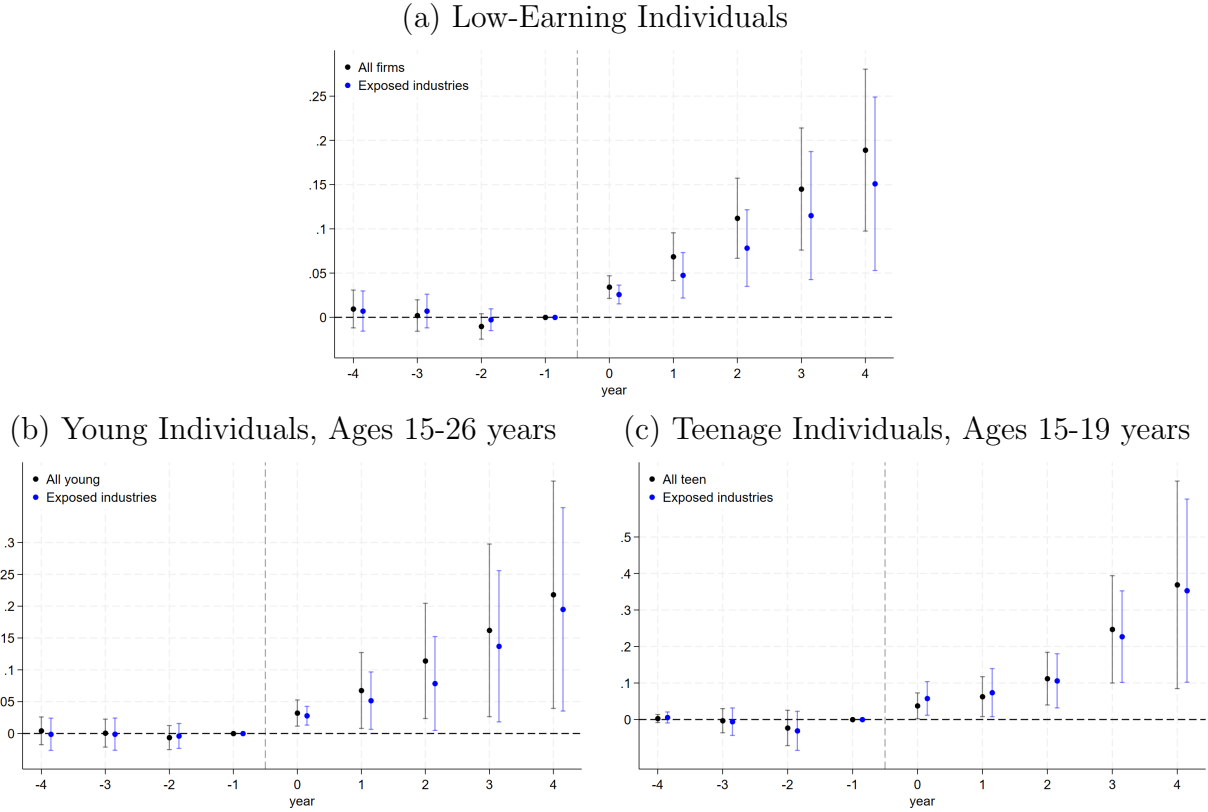
Table I.23: Robustness - Individual Panels: Employment Effects

	Across regression controls				Regression adjusted	N
	(1)	(2)	(3)	(4)		
All low-earning	0.0008 (0.0081)	0.0006 (0.0076)	0.0003 (0.0070)	-0.0024 (0.0068)	0.0006 (0.0087)	2,545,172
Low-earning exposed industries	0.0025 (0.0069)	0.0021 (0.0070)	0.0053 (0.0068)	0.0038 (0.0066)	0.0051 (0.0080)	644,636
All young	0.0123*** (0.0030)	0.0134*** (0.0037)	0.0186*** (0.0042)	0.0139*** (0.0038)	0.0106* (0.0055)	1,235,008
All teen	0.0146*** (0.0032)	0.0141*** (0.0035)	0.0170*** (0.0035)	0.0139*** (0.0045)	0.0099** (0.0050)	414,834
<u>Controls:</u>						
Baseline industry		X	X	X	X	
Age and age ²			X	X	X	
local market characteristics				X	X	

Note: This table presents the estimated effects of the minimum wage increases on employment for baseline low-earning workers and young individuals, using various sets of controls and an alternative estimation strategy. In the first four columns, estimates are from linear probability model regressions of Equation (2) where the dependent variable is an indicator if the individual receives a W-2 in year s . Reported coefficients are estimates of the average change in employment probability for each group between the base year prior to the policy change $s - 1$ and four years hence $s + 4$. Col. (1) shows results including only controls for individual-cohort and cohort-by-year fixed effects, Col. (2) adds controls for baseline industry, col (3) adds baseline age and age-squared, Col. (4) adds baseline quintiles of county density and quintiles of county employment rates. All controls are flexibly interacted with year. Standard errors clustered at the state-by-cohort level. The last column shows results using a “regression adjusted” (RA) estimator, described in detail in Appendix J. The RA estimation uses the full set of controls and standard errors estimated using clustered bootstrapping with resampling at the state-cohort level.

I.2. Percent changes

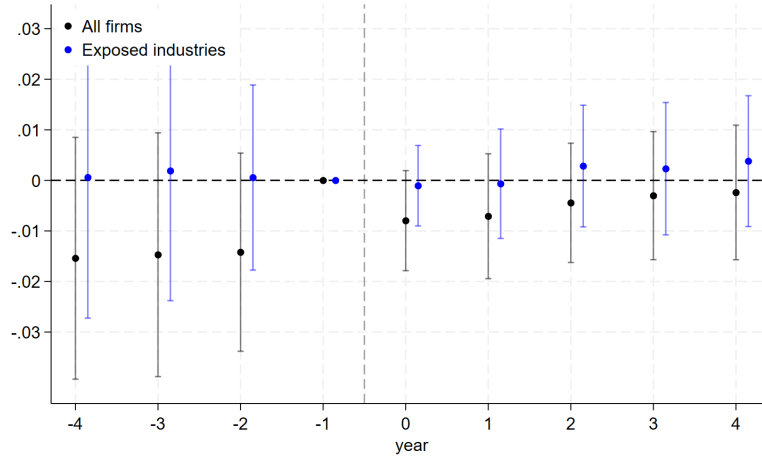
Figure I.11: Impact of Minimum Wage Increases on Earnings Trajectories - Percent Changes



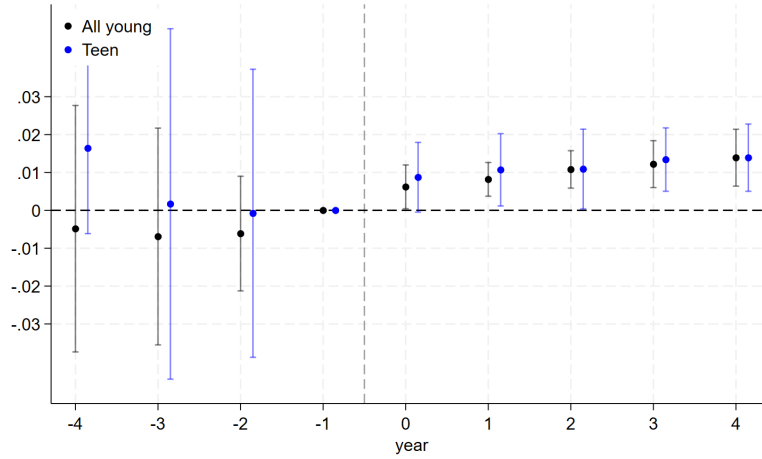
Note: This figure demonstrates the impact of minimum wage increases on the evolution of average earnings for low-earning and young workers. Plotted coefficients are estimates of β_s from Equation (2) where the dependent variable is annual individual earnings scaled by baseline ($t-1$) annual earnings. Thus, the coefficients represent estimated percent change in annual earnings due to the minimum wage increase. Panel (a) plots the difference in the earnings trajectories of low-earning workers ($<20,000$ but >0 in year $s-1$, and $<25,000$ in year $s-2$) in treatment relative to control states. Trajectories are plotted separately for all low-earning individuals and those employed in highly exposed industries in base year $s-1$. Panel (b) plots the impact on earnings trajectories for young individuals ages 15-26 in base year $s-1$. Panel (c) focuses on individuals who were teen age, 15-19 in base year $s-1$. In addition to individual-cohort and cohort-by-year fixed effects, the specifications include baseline age and age squared, quintiles of county density and quintiles of county employment rates – all flexibly interacted with year to allow for differential time trends. For each point estimate the 95% confidence interval is marked with standard errors clustered at the state-by-cohort level.

Figure I.12: Individual Employment, Percent Changes

(a) Low Earning Worker Panel



(b) Young Individual Panel

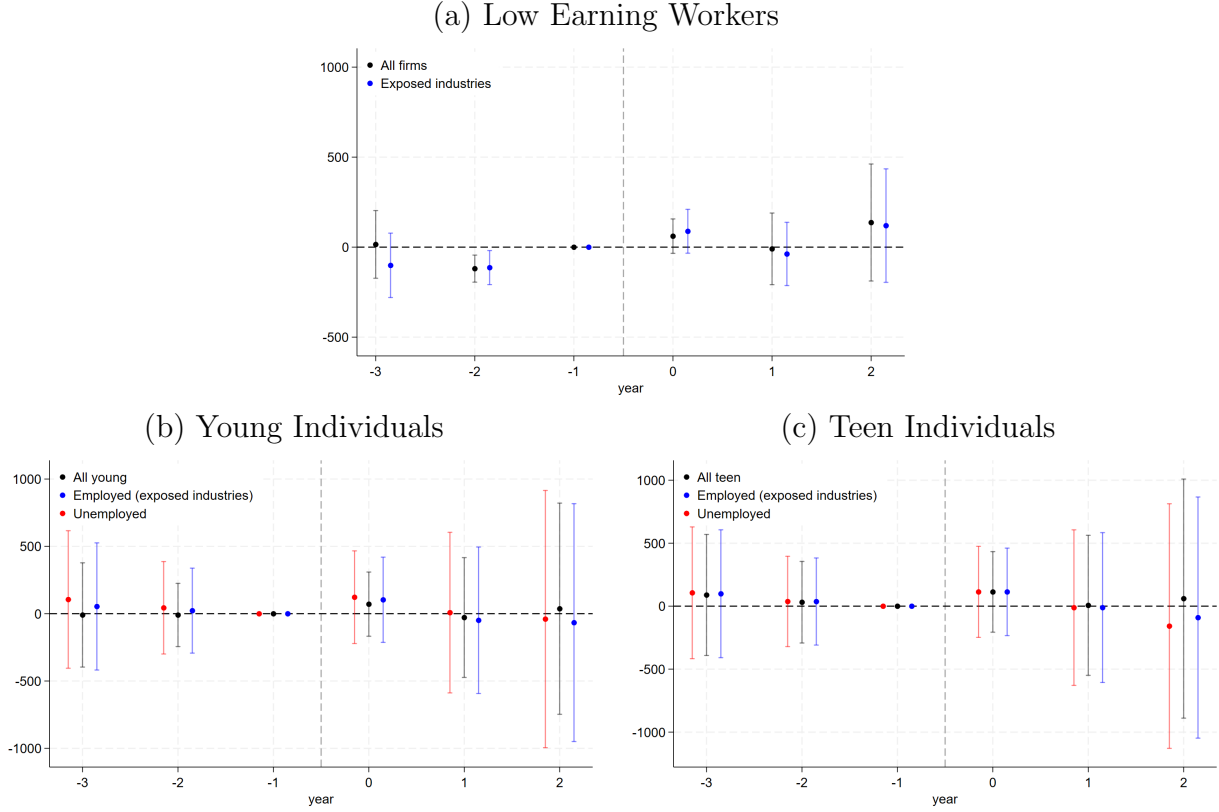


Note: This figure demonstrates the impact of minimum wage increases on the evolution of employment probabilities for low-earning and young workers. Plotted coefficients are estimates of β_s from Equation (2) where the dependent variable is an indicator for being employed in year t . Panel (a) plots the difference in the earnings trajectories of low-earning workers ($<20,000$ but >0 in year $s - 1$, and $<25,000$ in year $s - 2$) in treatment relative to control states. Trajectories are plotted separately for all low-earning individuals and those employed in highly exposed industries in base year $s - 1$. Panel (b) plots the impact on earnings trajectories for young individuals, ages 15-26, and teens in base year $s - 1$. In addition to individual-cohort and cohort-by-year fixed effects, the specifications include baseline age and age squared, quintiles of county density and quintiles of county employment rates – all flexibly interacted with year to allow for differential time trends. For each point estimate the 95% confidence interval is marked with standard errors clustered at the state-by-cohort level.

I.3. Placebo Tests

Although visual inspection of pre-trends in each of the individual analyses presented in the main results suggests that individuals in treatment and control states were on similar trajectories prior to the minimum wage increases, we also conduct a formal placebo test to examine

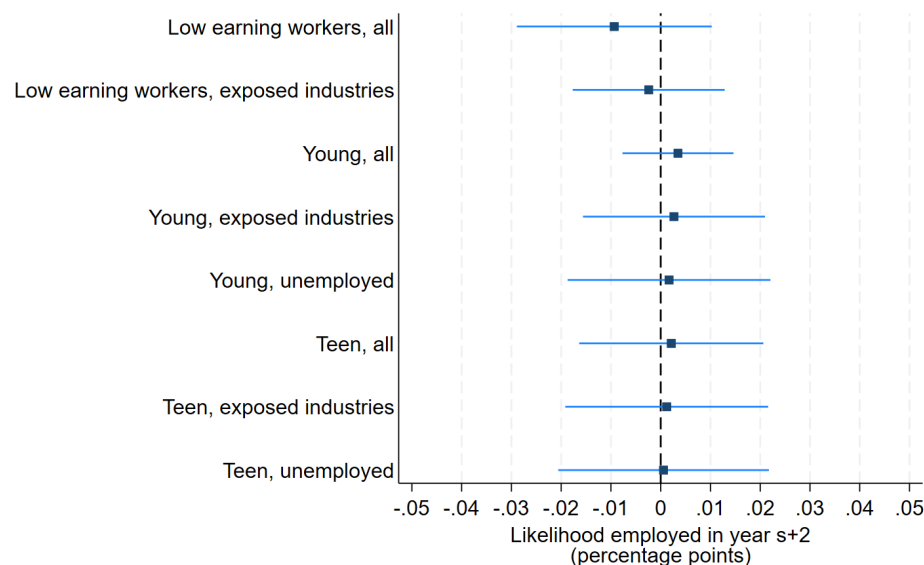
Figure I.13: Placebo Test - Individual Earnings “Effects” for Placebo Treatment Year



Note: This figure shows results of a placebo test where we estimate differences in individual earnings between treatment and control states around a placebo minimum wage increase in year 2010, (i.e. at least three years prior to the first minimum wage increases). This mirrors the main estimates and estimation strategy discussed in Section 5 presented in Figure 7, but the base year in Equation (2) is 2009 for all states, and the low-earning worker and young individual panels are sampled using base year 2009 characteristics. In addition to individual-cohort and cohort-by-year fixed effects, the specifications include baseline age and age squared, quintiles of county density and quintiles of county employment rates – all flexibly interacted with year to allow for differential time trends. For each point estimate the 95% confidence interval is marked with standard errors clustered at the state-by-cohort level.

the comparability of treatment and control groups. Appendix Figure I.13 and Appendix Figure I.14 show the results of placebo tests where we estimate differences in individual earnings and employment in treatment and control states around a placebo minimum wage increase in 2010, which is three years before any of our true policy changes. All estimates show no impact of the placebo event.

Figure I.14: Placebo Test - Individual Employment “Effects” for Placebo Treatment Year

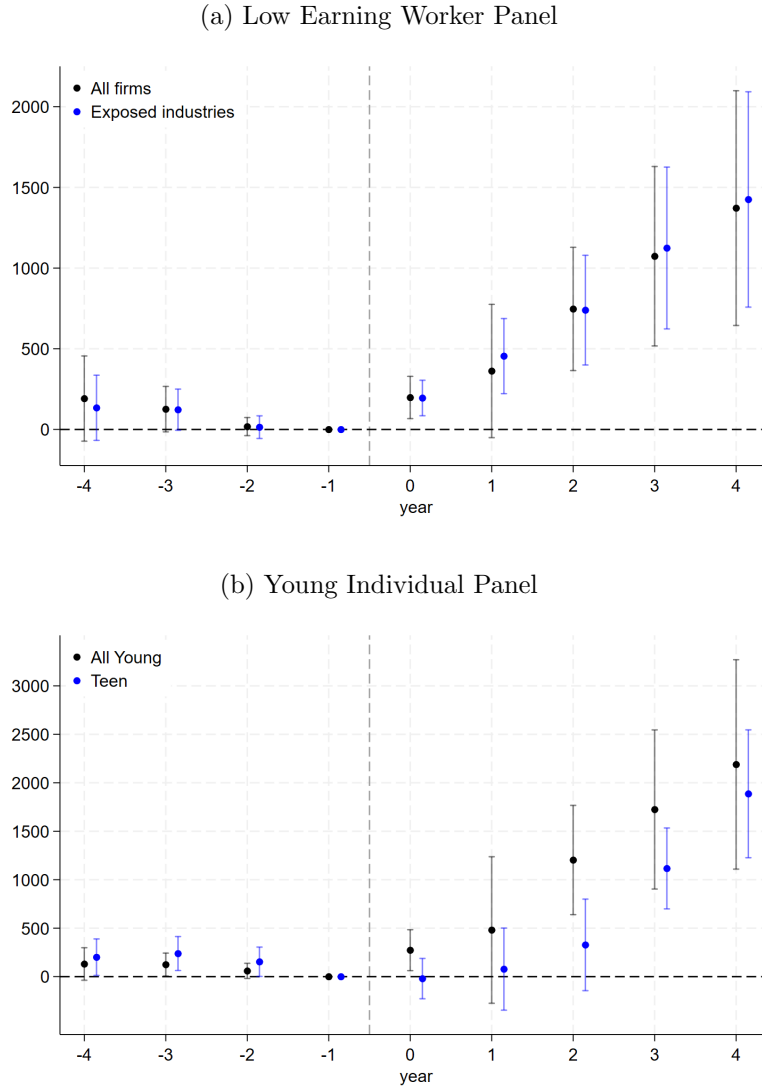


Note: This figure shows results of a placebo test where we estimate differences in individual employment probabilities between treatment and control states around a placebo minimum wage increase in year 2010, (i.e. at least three years prior to the first minimum wage increases). This mirrors the main estimates and estimation strategy discussed in Section 5 presented in Figure 8, but the base year in Equation (2) is 2009 for all states, and the low-earning worker and young individual panels are sampled using base year 2009 characteristics. Plotted coefficients are estimates of the average change in employment probability for each group between the placebo base year, 2009, and three years after the placebo minimum wage increase, 2012. In addition to individual-cohort and cohort-by-year fixed effects, the specifications include baseline age and age squared, quintiles of county density and quintiles of county employment rates – all flexibly interacted with year to allow for differential time trends. For each point estimate the 95% confidence interval is marked with standard errors clustered at the state-by-cohort level.

I.4. Excluding California

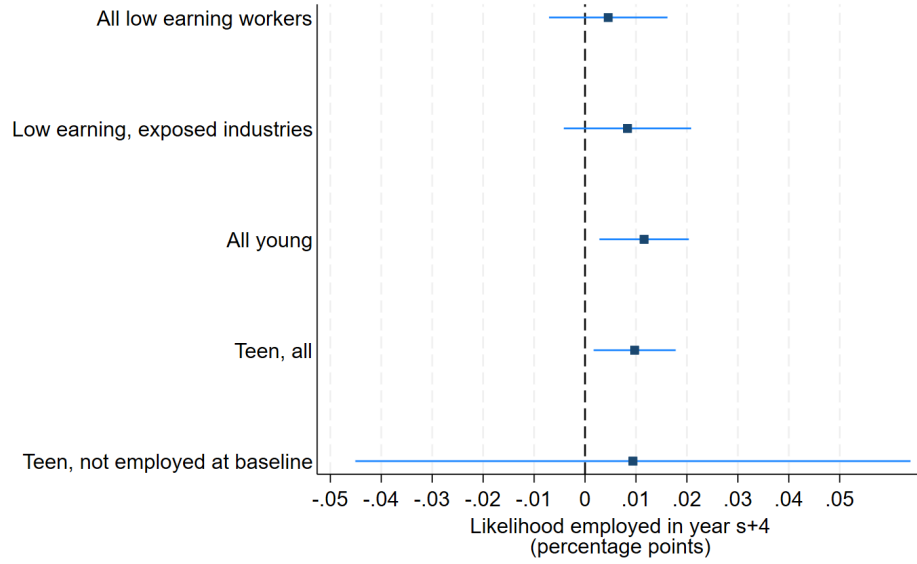
Appendix Figure I.15 and Figure I.16 show the individual earnings and employment impacts of minimum wage changes when excluding California as a treatment state in the analysis.

Figure I.15: Individual Earnings - Robustness, Excluding California



Note: This figure demonstrates the impact of minimum wage increases on the evolution of average earnings for low-earning and young workers when excluding California (CA) as a treatment state in the analysis. This tests robustness to dropping the largest state in the sample. CA accounts for almost 32% of low-earning workers in the main treatment group sample, $\approx 30\%$ of low-earning workers in highly exposed industries, $\approx 31\%$ of young individuals and $\approx 24\%$ of teens. Plotted coefficients are estimates of β_s from Equation (2) where the dependent variable is annual earnings. Panel (a) plots the difference in the earnings trajectories for all low-earning individuals and those employed in highly exposed industries in base year $s - 1$. Panel (b) plots the impact on earnings trajectories for young individuals ages 15-26 in base year $s - 1$ separately by base year employment status. In addition to individual-cohort and cohort-by-year fixed effects, the specifications include baseline age and age squared, quintiles of county density and quintiles of county employment rates – all flexibly interacted with year to allow for differential time trends. For each point estimate the 95% confidence interval is marked with standard errors clustered at the state-by-cohort level.

Figure I.16: Individual Employment - Robustness, Excluding California

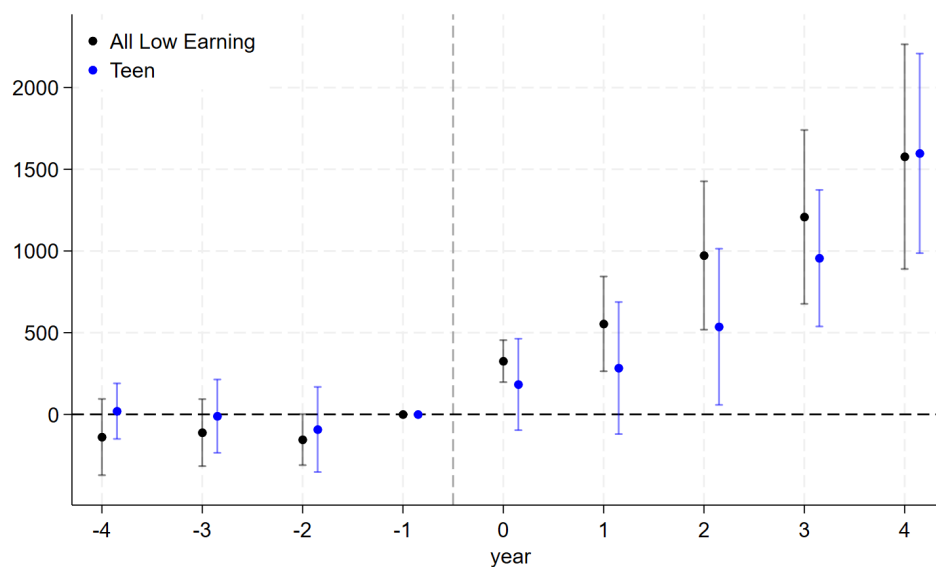


Note: This figure demonstrates the impact of minimum wage increases on employment for low-earning and young workers when excluding California (CA) as a treatment state in the analysis. This tests robustness to dropping the largest state in the sample. CA accounts for almost 32% of low-earning workers in the main treatment group sample, $\approx 30\%$ of low-earning workers in highly exposed industries, $\approx 31\%$ of young individuals and $\approx 24\%$ of teens. Estimates are from linear probability model regressions of Equation (2) where the dependent variable is an indicator if the individual receives a W-2 in year s . Plotted coefficients are estimates of the average change in employment probability for each group between the base year prior to the policy change $s-1$ and four years hence $s+4$. The top two estimates are from the low-earning worker panel and the lower three are from the panel of young individuals. In addition to individual-cohort and cohort-by-year fixed effects, the specifications include baseline age and age squared, quintiles of county density and quintiles of county employment rates – all flexibly interacted with year to allow for differential time trends. For each point estimate the 95% confidence interval is marked with standard errors clustered at the state-by-cohort level.

I.5. Individuals Employed by Independent Businesses at Baseline

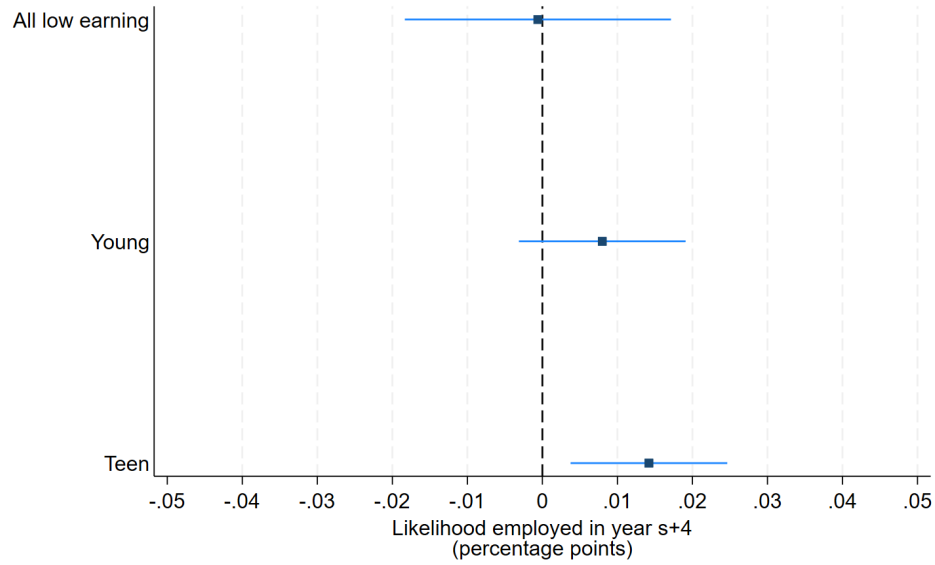
Appendix Figure I.17 and Appendix Figure I.18 examine the earnings trajectories and employment impacts of minimum wage increases for low-earning workers employed by independent businesses operating in highly exposed industries. The figures examine low-earning workers employed by highly exposed independent business overall, as well subgroups by age at baseline.

Figure I.17: Individual Earnings - Workers at Independent Businesses at Baseline



Note: This figure demonstrates the impact of minimum wage increases on the evolution of average earnings for workers at highly exposed independent businesses at baseline. Plotted coefficients are estimates of β_s from Equation (2) where the dependent variable is annual earnings across all jobs. The first series plots the difference in the earnings trajectories for low-earning workers (earning $<20,000$ but >0 across all jobs in year $s - 1$, and $<25,000$ in year $s - 2$) in highly exposed independent businesses at baseline. The second series shows estimates is for teen (15-19 years old) workers at these firms at baseline. In addition to individual-cohort and cohort-by-year fixed effects, the specifications include baseline age and age squared, quintiles of county density and quintiles of county employment rates – all flexibly interacted with year to allow for differential time trends. For each point estimate the 95% confidence interval is marked with standard errors clustered at the state-by-cohort level.

Figure I.18: Individual Employment - Workers at Independent Businesses at Baseline



Note: This figure demonstrates the impact of minimum wage increases on employment four years after the minimum wage increases for workers at highly exposed independent businesses at baseline. Estimates are from linear probability model regressions of Equation (2) where the dependent variable is an indicator if the individual receives a W-2 in year s . Plotted coefficients are estimates of the average change in employment probability for each group between the base year prior to the policy change $s - 1$ and four years hence $s + 4$. The first estimate presents outcomes for low-earning workers (earning $<20,000$ but >0 across all jobs in year $s - 1$, and $<25,000$ in year $s - 2$) in highly exposed independent businesses at baseline. The second estimate is for young (15-26 years old) workers at these firms at baseline; the third is for teenagers (15-19 years old) at these firms at baseline. In addition to individual-cohort and cohort-by-year fixed effects, the specifications include baseline age and age squared, quintiles of county density and quintiles of county employment rates – all flexibly interacted with year to allow for differential time trends. For each point estimate the 95% confidence interval is marked with standard errors clustered at the state-by-cohort level.

J. Regression Adjusted Estimators

Our primary regression specifications to estimate treatment effects are linear regressions of the form represented in Equations (1) and (2). Roth et al. (2023) show that even when the identifying parallel trends assumption holds conditional on control variables X_j , linear regressions can lead to inconsistent estimates of the treatment effect in DiD estimators if treatment effects are heterogeneous in X_j . While our use of non-parametric controls mitigates this concern, we also present estimates using “regression adjusted” estimators of the treatment effect which provides consistent estimates under weaker homogeneity assumptions than those required with linear regression (Roth et al. (2023)).

Regression adjustment is a two-step approach to estimating treatment effects. First, separate models are fit regressing the outcome on control variables separately for the treatment and control groups. To estimate the average treatment effect (ATE), the differences in the average predicted values for the treatment and control groups give the ATE. To estimate the average treatment effect on the treated (ATT), the predicted values for the treatment group from both the control and treatment regressions are used, (i.e. the predicted control values are evaluated using the observed covariates of the treatment group).

We estimate the treatment effects in a long differences specification, comparing outcomes of treatment and control groups from base year $s-1$ to post year $s+4$. The first-step equation is:

$$y_{jct} = \alpha + \Gamma_s X_{jc} + I(post_{s=t}) + I(post_{s=t}) \times \Gamma_s X_{jc} + \delta_{ct} + \nu_{jct} \quad (J.2)$$

where $I(post_{s=t})$ is an indicator for event year $s+4$. This model is run separately for the treatment and control groups. The ATE is estimated by taking the mean difference-in-differences in the predicted outcomes for the treatment and control group. Define $\hat{Y}_{T,t}(T)$ as the average predicted outcomes where: subscript $T \in 0, 1$ indexes that values are predicted from the first step regression run on the control group (0) or treatment group (1), $t \in pre, post$ indexes that the event year of the predicted values, and the T in parentheses indicates that the predicted values are evaluated at the observed covariates of group T. Then, the ATE and ATT are estimated as:

$$ATE = (\hat{Y}_{1,post}(1) - \hat{Y}_{1,pre}(1)) - (\hat{Y}_{0,post}(0) - \hat{Y}_{0,pre}(0))$$

$$ATT = (\hat{Y}_{1,post}(1) - \hat{Y}_{1,pre}(1)) - (\hat{Y}_{0,post}(1) - \hat{Y}_{0,pre}(1)).$$

We estimate these treatment effects separately for each outcome of interest. The ATT estimates for the firm results are reported in Table J.24 and the ATT estimates for the individual results are reported in Tables I.22 and I.23. Standard errors are obtained using clustered bootstrapping, where the first and second steps are jointly bootstrapped and resampling is done at the state-cohort level.

Table J.24: Robustness - Regression Adjusted Estimators

	Outcomes scaled by baseline revenues, $s-1$ to $s+4$				
	Revenue	Wage bill	COGS	Profits	Employment
All exposed	0.0351*	0.0143***	0.0132**	0.0005	-1.44**
	(0.0210)	(0.0036)	(0.0062)	(0.0041)	(0.684)

Note: This table presents estimates of the effect of the effect of minimum wage increases on firm outcomes using a “regression adjusted” (RA) estimator. The details of the two-step RA estimation are discussed in Section J. The estimation uses the same set of controls as in the main specification, cohort-by-year fixed effects, categories of baseline firm size (# of workers), deciles of baseline firm value-added, baseline two-digit industry, quintiles of county density, and quintiles of county employment rates – all interacted with year. The estimates are for differences between in outcomes for firms in treatment and control states four years after the minimum wages increases (year $s+4$), relative to pre-reform year $s-1$. Revenue, wage bill, COGS and profits are scaled by baseline revenue and weighted by log baseline revenue. Employment is estimated in levels. Standard errors are obtained using clustered bootstrapping, where the first and second steps are jointly bootstrapped and resampling is done at the state-cohort level.

K. Additional Individual Transition Analysis

To augment the transition analysis by firm type reported in the main results in Table 6, Appendix Table K.25 reports employment patterns four years after the minimum wage increases by size of the destination firm, and Appendix Table K.26 examines employment patterns over a shorter, two-year, horizon.

Panel A of Appendix Table K.25 reports differences in employment rates across the firm size distribution between low-earning workers in treatment and control states. Employment patterns are reported for all workers and just for those who switch employers (Movers). Low-earning workers are more likely to be employed by the largest employers after the minimum wage increase, with worker transitions away from the smallest firms in highly exposed industries and toward the largest employers outside of highly exposed industries driving the pattern. Panel B reports the results from our repeated cross-section analysis of teenagers (working and not). Teens are also less likely to be employed at the smallest firms in highly exposed industries.

Appendix Table K.26 repeats the analysis of Table 6 and Table K.25 using a shorter two-year horizon rather than the five-year horizon used in the other tables. The transition patterns we discuss over the five-year horizon are largely evident just two years after the minimum wage increases.

Table K.25: Employment Patterns of Low-Earning Workers and Teenagers

Panel A: Low Earning Workers Panel

	Firm size quartile (exposed industries)			
	Q1	Q2	Q3	Q4
<u>All workers</u>				
All industries	-0.0019** (0.0006)	-0.0009 (0.0007)	-0.00004 (0.0011)	0.0107*** (0.0038)
Exposed industries	-0.0088*** (0.0030)	-0.0050 (0.0033)	-0.0050 (0.0044)	0.0248* (0.0125)
<u>Movers</u>				
All industries	-0.0095** (0.0008)	-0.0016* (0.0008)	0.0003 (0.0011)	0.0075** (0.0036)
Exposed industries	-0.0103** (0.0041)	-0.0074* (0.0040)	-0.0064 (0.0048)	0.0176 (0.0128)

Panel B: Teens, repeated cross-section

	Firm size quartile (exposed industries)			
	Q1	Q2	Q3	Q4
All Teens	-0.0022** (0.0010)	-0.0031*** (0.0010)	0.0005 (0.0015)	-0.0004 (0.0051)
Employed teens	-0.0030* (0.0016)	-0.0049*** (0.0017)	0.0025 (0.0021)	-0.0003 (0.0076)

Note: The table above reports employment patterns for individuals from our low-earning panel and repeated cross-sections of teenagers. Employment patterns are tracked by firm size quartile, with size is defined by total labor payments as detailed in Appendix L.3. Panel A describes the impact of minimum wage increases on employment transitions between years $s - 1$ and $s + 4$ for a panel of low-earning workers. These workers are all employed at baseline in either exposed or unexposed industries and the estimates report the change in their transition patterns relative to similar workers in untreated states. Each row describes transition patterns conditional on the industry in which low-earning workers were employed in the base year. Movers columns focus on transition patterns of workers no longer employed by their baseline ($s - 1$) employer in year $s + 4$. Estimates are from a linear probability model specification of Equation (2) where the dependent variable is an indicator for a worker being employed at a firm in a given size quartile in year $s + 4$. The lower panel describes employment patterns for randomly sampled repeated cross-sections of teenagers. Section L.4 details the construction of these data. The reported estimates convey five-year changes in average employment rates at firms in different size quartiles between years $s - 1$ and $s + 4$. All standard errors are clustered at the state-cohort level. Regression N's by row: Panel A: 3,222,100; 987,388; 2,603,155 ; 842,201. Panel B: 5,338,697; 2,370,428.

Table K.26: Employment Patterns of Low-Earning Workers and Teenagers - Short Differences (2 year)

Panel A: Low Earning Workers Panel

Baseline industry	retain	exposed industry	All workers				Movers			
			exposed industry	exposed industry	unexposed industry	unexposed industry	exposed industry	exposed industry	unexposed industry	unexposed industry
			independent business	Large C-corp.	independent business	large C-corp.	independent business	large C-corp.	independent business	large C-corp.
Exposed industries	0.0272*** (0.0051)	0.0110** (0.0038)	-0.0096** (0.0043)	0.0191*** (0.0057)	0.0015 (0.0032)	-0.0068*** (0.0026)	-0.0222*** (0.0079)	0.0167** (0.0068)	0.0026 (0.0053)	-0.0109*** (0.0040)
Unexposed industries	0.0182*** (0.0065)	-0.0080* (0.0041)	-0.0024 (0.0025)	-0.0034*** (0.0013)	0.0005 (0.0053)	0.0023 (0.0042)	-0.0020 (0.0039)	-0.0052** (0.0022)	0.0016 (0.0110)	-0.0025 (0.0048)
			All workers				Movers			
			Firm size quartile (exposed industries)				Firm size quartile (exposed industries)			
			Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
All industries			-0.0015*** (0.0005)	-0.0006 (0.0005)	-0.00004 (0.0009)	0.0096*** (0.0033)	-0.0020** (0.0007)	-0.0011 (0.0007)	-0.0009 (0.0009)	0.0062* (0.0037)
Exposed industries			-0.0060*** (0.0023)	-0.0041 (0.0025)	-0.0059 (0.0038)	0.0274** (0.0112)	-0.0083** (0.0039)	-0.0065* (0.0034)	-0.0099** (0.0046)	0.0201 (0.0126)

Panel B: Teens, repeated cross-section

	Earnings (\$)	Employed	exposed industry	unexposed industry	exposed industry	exposed industry	Firm size quartile (exposed industries)			
			any	any	independent business	large C-corp.	Q1	Q2	Q3	Q4
All Teens	57 (71)	-0.0041 (0.0060)	-0.0080 (0.0052)	0.0040* (0.0021)	-0.0100*** (0.0029)	0.0026 (0.0022)	-0.0007 (0.0009)	-0.0022*** (0.0006)	-0.0001 (0.0011)	-0.0050 (0.0038)
Employed teens	191 (132)	. (.)	-0.0079 (0.0064)	0.0079 (0.0064)	-0.0164*** (0.0042)	0.0083** (0.0036)	-0.0007 (0.0014)	-0.0039*** (0.0011)	0.0009 (0.0018)	-0.0042 (0.0053)

Note: The table above reports employment patterns for individuals from our low-earning panel and repeated cross-sections of teenagers. Employment patterns are tracked by industry, firm type, and size quartile (with size defined by total labor payments). Panel A describes the impact of minimum wage increases on employment transitions between years $s - 1$ and $s + 2$ for a panel of low-earning workers. These workers are all employed at baseline in either exposed or unexposed industries and the estimates report the change in their transition patterns relative to similar workers in untreated states. Each row describes transition patterns conditional on the industry in which low-earning workers were employed in the base year. Movers columns focus on transition patterns of workers no longer employed by their baseline ($s - 1$) employer in year $s + 2$. Estimates are from a linear probability model specification of Equation (2) where the dependent variable is an indicator for a worker being employed at a given type of firm in year $s + 2$. The lower panel describes employment patterns for randomly sampled repeated cross-sections of teenagers. Section L.4 details the construction of these data. The reported estimates convey three-year changes in average employment rates in different industries and firm types measured between years $s - 1$ and $s + 2$. All standard errors are clustered at the state-cohort level. Regression N's by row: Panel A: 987,388; 1,485,775; 842,201; 1,335,715. Panel B: 5,338,697; 2,370,428.

L. Administrative Data and Empirical Methods

L.1. Linked Firm-Worker Data: Details and Variable Definitions

The spine of our firm-worker dataset is a 100% sample of all active pass-through firms in treatment and control states from 2010-2019.⁵⁷ From the firm income tax returns we keep information on firm revenues, cost of goods sold (COGS), value-added (gross profits, or revenues-COGS), net business income (firm profits or losses), all business deductions (deductible expenses), industry (6 digit NAICS codes), state and zipcode. The line item “other deductions” includes “total allowable trade or business deductions that aren’t deductible elsewhere” (quoted from the instructions of Form 1120S).⁵⁸ We define non-labor “intermediate costs” as the sum of COGS and the “other deductions” line items and firm “variable costs” as the wage bill plus non-labor intermediate costs.

To each active firm we link all workers in all years, defined as those receiving a W-2 with non-zero wage and salary income from that firm. For each worker, we keep annual wage and salary income from the W-2. We separately link information on each worker’s age. Using the information from worker W-2s, we define the total wage bill for each firm-year (the sum of all W-2 wage and salary amounts) and the number of employees in the year (total number of distinct workers receiving wages from the firm in a year). We also use these data to estimate the changes in the distributions of worker characteristics by including the number of workers in different age or earnings groups at each firm and the share of the wage bill going to those within earnings groups (see Figures 2 and 3).

Further, we use earnings information to define a subset of persistently low-earning workers that are more likely to be exposed to minimum wage changes. We define low-earning workers in the base year, $s-1$, based on their earnings history from year $s-2$ to year s to isolate persistently low earning workers as opposed to those with idiosyncratically low earnings in a given year. We define low-earning workers as those with total wage and salary earnings from all jobs that are less than 40 hours per week for 52 weeks at the minimum wage in each year from year $s-2$ to year s . We use this information to define the number of low-earning workers at each firm at baseline and the share of the firm’s wage bill and variable costs associated with earnings paid to low-earning workers. We note that while this group should contain essentially all minimum wage workers, it may also include consistently part-time workers that earn wages above the minimum wage.

Pass-throughs are required to file a Schedule K-1 on behalf of each owner, which reports the owner’s share of firm income in each year. To identify owners, we match each firm with all filed Schedule K-1 reports. Owners can be active, shareholder-employees, or passive owners. Firm owners earn annual profits or losses from their businesses and active owners pay themselves wage and salary income as well. We exclude active owners when calculating the number of workers at the firm and the firm’s annual wage bill. We define total owner income as the sum of net business income and the wages and salaries paid to active owners.

⁵⁷Pass-through firms are partnerships and S-corporations that file income tax return forms 1065 and 1120S, respectively. Active is defined as having non-zero net income in a given year.

⁵⁸Some of the eligible deductions listed on the instructions of the form include: Certain business start-up and organizational costs, insurance premiums, legal and professional fees, supplies used and consumed in the business, and utilities.

L.2. Analysis Samples

The various samples of firms used in throughout the analysis are summarized in Table L.27. The full sample of firms contains all active pass-throughs in treatment and control states in the 9 years around each cohort year s , i.e., from $s-4$ to $s+4$. The full sample includes approximately 2.53 million firms per year. The main firm results presented in Sections 4.1 to 4.3 use a balanced panel of firms in highly exposed industries active in each year $s-4$ to $s+4$ to assess minimum wage increases impact operating firms. The balanced panel includes 66,172 treatment firms and 68,802 control firms per year. As robustness, we use an unbalanced panel of firms in highly exposed industries that are active in year $s-1$ but may enter after year $s-4$ or exit before $S+4$ (Appendix Table E.7). This unbalanced panel includes 125,431 treatment and 130,686 control firms in year $s-1$. As a placebo test, we examine the effects of the minimum wage change on firms in non-highly exposed industries (Figure H.8). This balanced panel includes 661,412 treatment and 721,168 control firms per year. The aggregate analyses of firm entry and exit in Sections 4.5 to 4.6 use the full set of active firms in highly exposed industries in each year. This sample includes approximately 130,000 treatment firms and 135,000 control firms per year.

Table L.27: Analysis Samples: Number of Firms

	All	Treatment	Control
All firms	2,528,909	1,206,465	1,322,444
Highly exposed industries	271,307	132,901	138,406
Highly exposed industries (Balanced)	134,974	66,172	68,802
Highly exposed industries (Unbalanced)	256,117	125,431	130,686
Non-highly exposed industries (Balanced)	1,385,973	661,412	721,168

Note: The table above reports sample sizes for the different firm analyses reported in the main text and appendix. Treatment and Controls states are defined in Table A.1.

L.3. Individual Datasets

Low-earning Worker panel. We construct a worker-level panel to study the effects of the minimum wage changes on low-earning workers. Our spine is all low-earning workers living in treatment and control states in base year, $s-1$. We define the state by the state of the address reported on the individual tax return (Form 1040) and define persistently low-earning workers as those with earning between \$1-20,000 from all employers in year $s-1$ and \$0-25,000 in year $s-2$. This spine includes low-earning workers in all industries and working for any type of firm including large corporations.

From this spine, we draw a 2% random sample. From this sample, we link on information for all jobs (Form W-2) for all individuals in each year from year $s-4$ to year $s+4$. From the W-2 we keep information on the wage and salary earnings and the employer’s (deidentified) Employer Identification Number (EIN). For each job, we collect information about the employer by linking the EIN to information on the industry, the business form (partnership, S-corporation, C-corporation, sole proprietorship or not-for-profit). Using these data, we create a balanced panel of workers, defining total earnings as the sum of wage and salary income from all W-2s in a given year and setting earnings as zero for those “not employed” in a given year, defined as those receiving no W-2s with positive earnings. We define the number of jobs as the number of W-2s from distinct employers in a given year. To assign industry and type of employer, we keep the two highest paying job in each year. Also, for each worker we collect information on the year of birth.

To estimate the firm size of each employer, we combine several pieces of information. First, we link on firm income tax returns from S-corporations (Form 1120S), Partnerships (Form 1065) and C-corporations (Form 1120) and keep information on the annual revenues. For many individuals working for large C-corporations, the EIN on the W-2 does not link directly to a corporate income tax return, either because it is a subsidiary with no direct filing requirement or because the EIN on the W-2 is associated with a payroll service provider rather than the corporation itself. To recover a consistent measure of firm size for all firms, including these large corporations, we collect information from Form 941 (Employer’s Quarterly Federal Tax Return), which reports information on total quarterly payrolls and number of employees for payroll tax purposes. For each firm, we average the values across the four quarters to get a measure of annual payroll and employment. We then rank all firms according to their position in the distributions of payroll and employment as a measure of relative firm size. We use quartiles of the distribution of total payroll to estimate differential transitions across the firms size distribution, using the individual LPM specification (2) where the outcome variable is an indicator for working at a firm in a given quartile of the size distribution (Appendix Table K.25).

Young and teen panel. We create a second individual panel focusing on young individuals - those with or without a job in the pre-reform year $s-1$. The sample frame is all individuals ages 15-26 in treatment in control states in year $s-1$. These are individuals who file tax returns on their own, receive a W-2 from an employment relationship, or are listed as dependents on a household return. From this frame, we take a 2% random sample of individuals in the baseline year.

For this spine of individuals, we link on information for all jobs (Form W-2) for all individuals in each year from year $s-4$ to year $s+4$. From the W-2 we keep information the

wage and salary earnings and the employer’s (deidentified) Employer Identification Number (EIN). Using these data, we create a balanced panel of workers, defining total earnings as the sum of wage and salary income from all W-2s in a given year and setting earnings as zero for those “not employed” in a given year, defined as those receiving no W-2s with positive earnings. we define the number of jobs as the number of W-2s from distinct employers in a given year. We perform separate analyses with this panel for those working at baseline and for those not employed (receiving no W-2s) at baseline.

L.4. Repeated Cross-Section of Teens

In Section 5.6, we analyze a repeated cross-section of teenagers in treatment and control states for years $s-4$ to $s+4$. To estimate effects on employment rates of concurrent teenagers in each year surrounding the minimum wage increases, we want a representative sample of teens in each year whether employed or not. To do so, we start with a sample frame of all individuals with birth years corresponding to 16-19 year olds in treatment and controls states in a given sample year. The year of birth data is populated for all individuals who filed an income tax return, were claimed as dependents on a parent or guardian’s tax return or received a W-2 information report at any point before by year 2023. This allows us to sample teens that may not have been employed in any given year, and may not have been employed yet by the sample year. Yet, administrative tax records are not designed for statistical analysis of employment rates or to make representative subsamples of U.S. populations, so this sample may not be perfectly representative of the population of teens, particularly since it will not contain teens that never appear on a tax return before 2023, which may be disproportionately those that never work or that are from low income families. For this reason, we conduct complementary analysis using the nationally representative American Community Survey (ACS) as described in Section 5.6 and Appendix N.

Starting from this frame of the available population of teenagers in treatment and control states in each year from event years $s-4$ to $s+4$, we draw a 2% random sample of teens in each year. For this sample, we link on information for all jobs (Form W-2) in that year. From the W-2 we keep information the wage and salary earnings and the employer’s (deidentified) Employer Identification Number (EIN). We define total earnings as the sum of wage and salary income from all W-2s in a given year and set earnings to zero for those “not employed” in a given year, defined as those receiving no W-2s with positive earnings. we define the number of jobs as the number of W-2s from distinct employers in a given year. Using the employer EIN we classify employers by their business form and size in the same way as was done for the low-earning panel as previously described.

L.5. Collapsed Data and Aggregate Effects

Sections 4.5 and 4.6 present aggregate analyses of the effects of the minimum wage on firm entrance and exit, and the effects on sector-level aggregates (where “sector” is defined as independent firms in highly exposed industries), respectively. To conduct these analyses in a well-identified causal framework, we use collapsed datasets to estimate aggregates.

Firm Entry and Exit. To estimate differential firm entry and exit, we create a collapsed dataset that allows us to estimate differential entry and exit rates while controlling for relevant market characteristics to approximate the firm-level regressions. Concretely,

we collapse the firm-level dataset by cohort, year, treatment and control states, industry, quintiles of county density, and quintiles of pre-period firm churn at the commuting zone level. The first controls mimic the market level controls from the firm analysis. The last control, for firm churn at the CZ-level is added to control for average pre-period differences in firm entrance and exit across treatment and control states. Pre-period entrance and exit rates are taken from years $s-4$ to $s-1$. Data are collapsed by these variables and counts of firms within each cell are retained, differentiating between “incumbent firms”, or those that operate in year t and operated in base year $s-1$ and “entrant firms”, those that operate in year t but did not operate in year $s-1$. The total number of firms operating in each cell is the sum of the incumbent and entrant firms in that cell.

Regressions are run on the collapsed data using a specification is similar to that described by Eq. 1:

$$y_{hct} = \alpha + \sum_{s=-4, s \neq -1}^4 (\beta_s treat_{hc} + \Gamma_s Z_{hc}) \times year_{s=t} + \delta_{ct} + \omega_{hc} + \nu_{hct} \quad (L.3)$$

where h indexes treatment or control firms within a given industry-by-county density quintile-by-CZ churn quintile. The outcome variables are scaled by baseline values within cohort and set of market characteristics, h , to estimate relative changes. Concretely, the outcome variables are total firms in hc in year t over the total number of firms in hc in year $s-1$, total number of entrant firms in hc in year t over the total number of firms in hc in year $s-1$, and total number of incumbent firms in hc in year t over the total number of firms in hc in year $s-1$. The regressions are weighted by the baseline number of firms in $hc, s-1$. This methodology allows us to estimate changes in the total number of firms operating, decomposed into entry and exit rates using a DiD framework similar to the main firm-level specifications.

For firm exit rates, which can be estimated using firm-level regressions, we validate the exit results using Specification (1) with the standard set of firm and market controls and the outcome variable being an indicator for the firm operating in year t . Appendix Figure H.10 shows the results of this exercise, which are very similar to the main results using the collapsed data, shown in Figure 6.

Aggregate Effects. We also use collapsed data to estimate the aggregate effects of the minimum wage increases on the sector of independent businesses in highly exposed industries. To do so, we create a simple state-by-industry dataset, collapsing firm-level income statement data by cohort, year, state and industry. We run state-level regressions of the form:

$$y_{rct} = \alpha + \sum_{s=-4, s \neq -1}^4 (\beta_s treat_{rc} + \Gamma_s \chi_{rc}) \times year_{s=t} + \delta_{ct} + \phi_{rc} + \nu_{rct} \quad (L.4)$$

where r indexes state, χ_{rc} is a state-by-cohort fixed effect and χ_{rc} are a vector of indicators for industry. Outcome variables are state-by-industry totals of a given outcome scaled by baseline ($s-1$) revenue within state-by-industry groups for comparison with the main firm-level results using the balanced panel. Regressions are weighted by baseline revenues.

M. Border Design

We provide additional analyses using a border county design similar to that advanced in [Card and Krueger \(2000\)](#), [Dube et al. \(2010, 2016\)](#) and [Allegretto et al. \(2017\)](#). To implement this design within our context, we use treatment states (T) and the control states (C) in our sample that border these states. Of the 19 treatment locations in our sample, 15 have bordering control states.⁵⁹ The state pairs that contain the counties in this analysis are: Arkansas (T) with Mississippi, Oklahoma, Tennessee and Texas (C); California (T) with Nevada (C); Washington, DC (T) with Virginia (C); Delaware (T) with Pennsylvania (C); Chicago (T) with Indiana (C); Massachusetts (T) with New Hampshire (C); Maryland with Pennsylvania and Virginia (C); Maine (T) with New Hampshire (C); Michigan (T) with Wisconsin and Indiana (C); Minnesota (T) with Wisconsin and North Dakota (C); Nebraska (T) with Wyoming and Kansas (C); New Jersey (T) with Pennsylvania (C); New York (T) with Pennsylvania (C); South Dakota (T) with North Dakota and Wyoming (C); and West Virginia (T) with Pennsylvania (C). Within these border states, we select firms in counties with centroids within 30 miles of the state border. When a county is within 30 miles of multiple borders, the firms in that county will have multiple observations - one observation per border pair ([Dube et al. \(2010, 2016\)](#)).

This design compares independent firms in highly exposed industries in states that raise the minimum wage to similar firms just across the border in a state that did not have a minimum wage increase in the period. The regression specification is similar to that used in the main analysis (equation (1)), but adding fixed effects for border pairs interacted with year:

$$y_{jct} = \alpha + \sum_{s=-4, s \neq -1}^4 (\beta_s treat_{jc} + \Gamma_s X_{jc} + \Psi_s I_{jc}) \times year_{s=t} + \delta_{ct} + \psi_{jc} + \nu_{jct} \quad (M.5)$$

where the variables are defined as in equation (1) with the addition of I_{jc} which represents border pair fixed effects.

The results of this analysis are presented in Table [M.28](#). The reported estimates are DiD coefficients for the differential change in the outcome variable for treatment firms relative to control firms from years $s - 1$ to $s + 4$ (i.e. β_4 from equation (M.5)). Panel A shows results for the main outcomes - revenues, wage bill, COGS and profits scaled by baseline revenues, as well as level employment and the estimated own-wage-elasticity (OWE).

While the magnitudes of the estimated effects on firm outcomes are somewhat attenuated, the qualitative results are very similar when using the border firms as when using the full sample. Revenues raise sufficiently offset the new higher wage and non-labor costs such that owner profits do not decrease. The estimated employment effects and corresponding OWEs are very similar to those estimated using the full sample as well.

⁵⁹Only Connecticut, Rhode Island, Hawaii and Alaska do not.

Table M.28: Effects of the Minimum Wage - Border Design

	Outcomes scaled by baseline revenues, $s-1$ to $s+4$					
	Revenue	Wage bill	COGS	Profits	Employment	OWE
All exposed	0.0116 (0.0076)	0.0086*** (0.0028)	0.0027 (0.0028)	-0.0006 (0.0020)	-1.369** (0.665)	-0.376 (0.271)
N	114,031	114,031	114,031	114,031	114,031	114,031

Note: Table M.28 presents results from the border design and regression specification described in Appendix M. The reported estimates are DiD coefficients for the differential change in the outcome variable for treatment firms relative to control firms from years $s - 1$ to $s + 4$ (i.e. β_4 from equation (M.5)). The first four columns show the estimated effects on revenues, wage bill, COGS and profits scaled by baseline revenues. The employment effect is estimated in levels and the OWE is estimated by dividing the log employment estimate by an estimate of the log change in average wages, with the standard error estimated using the delta method. The border design uses firms in counties with centroids within 30 miles of the state border. Outcomes are winsorized at the 1st and 99th percentiles. In addition to firm-cohort and cohort-by-year fixed effects, and border-pair fixed effects interacted with year, the specifications include categories of baseline firm size (# of workers), deciles of baseline firm value-added, baseline two-digit industry, quintiles of county density, and quintiles of county employment rates – all flexibly interacted with year to allow for differential time trends. For the first four outcomes, regressions are weighted by log base year revenues. standard errors are clustered at the state-by-cohort level.

N. Earnings and Employment Analysis with Public Data

To complement the individual-level analysis results reported in Section 5, we conducted a similar analysis using publicly-available data from the American Community Survey (ACS). The ACS provides nationally representative cross-sections of the United States population by surveying approximately 295,000 addresses monthly.

As a representative survey of the U.S. population, the ACS is well-suited for providing estimates of the earnings and employment impacts of minimum wage policies. Importantly, these data include individuals who may not have recently filed a tax return — a necessary condition for being included in our linked panel drawn from IRS tax data. This is particularly useful when examining the impact of minimum wage increases on young individuals who are naturally less likely to have filed a return. While our main analysis with the tax data tries to address this issue by including individuals claimed as dependents in prior years, the ACS is designed to capture these and other non-workers. Further, unlike the tax data which report earnings, the ACS captures hours worked, an important dimension of the labor market impacts of minimum wages.

For the ACS analysis, we use IPUMS Amercian Community Survey data for years 2007 to 2020 (Ruggles et al., 2024). Our outcomes of interest are employment, usual hours worked and wage and salary income. To determine whether an individual was employed in a highly exposed industry, we compared the ACS industry code to our list of highly exposed industries based on Current Population Survey data (see Appendix Table B.3). We also use information on individual age, restricting our analysis to individuals ages 16 to 60 years old.

Individuals are classified into treatment and control groups according to their state of residence (see Appendix Table A.1). In the case of Illinois, individuals residing outside of Chicago are classified as control while those residing in Chicago are considered treated. Treated individuals are assigned to one of four cohorts based on the year their state of residence increased its minimum wage. These cohorts are also detailed in Appendix Table A.1.

Our regression specification follows:

$$y_{ict} = \alpha + \sum_{s=-4, s \neq -1}^4 (\beta_s treat_{ic} + \Gamma_s V_{ic}) \times year_{s=t} + \delta_{ct} + \nu_{ict} \quad (\text{N.6})$$

where, $treat_{ic}$ is a dummy equal to one if the individual resides in a state that raised its minimum wage, V_{ic} is a vector of control variables, δ_{ct} represents cohort-by-year fixed effects, and ν_{ict} is the error term. Note that this specification does not include the individual-cohort fixed effects of our main analysis as the ACS data comprise cross-sections rather than a panel. Like in the main analysis, the construction of the cohorts will mean that individuals in control states may appear in multiple cohorts. We avoid double-counting by clustering at the state-cohort level as described below.

The key parameters of interest are the β_s , which trace the impacts of the minimum wage increases on employment and earnings measures, averaging across the four cohorts. In Appendix Table N.29 below, we present estimates for four age groups: prime-age (16 to 60 years old), teenagers (16 to 19 years old), young individuals (16 to 26 years old), and young adults (20 to 26 years old).

Using the ACS data, we find no evidence of any impact on employment rates or employment in highly exposed industries for any of the age groups, either immediately after the minimum wage increase or four years after. Reported coefficients represent percentage point differences, so that, for example, the estimate for the prime-age subgroup indicates an insignificant 0.01 percentage point increase in employment. Usual hours worked per week are also unaffected. However, earnings are higher following the minimum wage increase. ACS data show that wage and salary income increases by \$1,411 for the all ages group. These gains are concentrated in older workers, as none of the three younger subgroups post statistically significant increases in wages and salary income.

These results help to put our main analysis using tax return data into context.

It is worth noting that for neither of the individual panels of our IRS data presented in Section 5 no longer condition on employment

Appendix Table N.30 reports the sensitivity of the above ACS results to different sets of controls. We focus on the full set of prime-age individuals ages 16 to 60 years. Starting without any controls, each row adds a new set of controls such that the final row corresponds to the prime-age estimates reported in the Appendix Table N.29. Local area characteristic controls are PUMA-level density and unemployment interacted with time fixed effects. Age controls add $age + age^2$ terms. The results are not qualitatively different across the specifications.

Table N.29: Employment and Earnings Effects of Minimum Wage Increase, ACS Data

	Prime age (16-60)	Young workers (16-26)	Teenager (16-19)	Young adults (20-26)
Employed	0.0001 (0.0033)	-0.0057 (0.0054)	-0.0118 (0.0076)	-0.0028 (0.0049)
Employed in an exposed industry	-0.0001 (0.0015)	-0.0004 (0.0037)	-0.0030 (0.0057)	0.0010 (0.0035)
Usual hours worked per week	0.01 (0.14)	-0.12 (0.23)	-0.21 (0.24)	-0.09 (0.25)
Wage and salary income	1,410.85 (274.99)	242.76 (176.84)	-76.66 (91.43)	374.76 (232.42)

Note: The table above reports estimates of Appendix Equation (N.6) using ACS data from 2007 to 2020 where the outcomes are 1) an employment dummy, 2) a dummy indicating a worker is employed in a highly exposed industry, 3) usual hours worked each week, and 4) total wage and salary income. Estimates report impacts four years after the policy change ($s+4$) relative to the year before the policy change ($s-1$). We estimate linear probability models for binary employment outcomes such that the coefficients measure percentage point differences. An individual is considered employed in an exposed industry if their ACS industry code corresponds with the list of highly exposed industries listed in Appendix Table B.3. Regressions are weighted by ACS person weights and standard errors are clustered at the state-cohort level.

Table N.30: Employment and Earnings Effects of Minimum Wage Increase: Specification Comparison, ACS Data

	No controls	With cohort-year fixed effects	With local area characteristic controls	With linear and nonlinear age controls
Employed	0.0045 (0.0039)	0.0037 (0.0035)	0.0005 (0.0034)	0.0001 (0.0033)
Employed in an exposed industry	0.0069 (0.0043)	0.0004 (0.0016)	-0.0002 (0.0015)	-0.0001 (0.0015)
Usual hours worked per week	0.22 (0.16)	0.17 (0.15)	0.03 (0.15)	0.01 (0.14)
Wage and salary income	1,387.57 (322.49)	1,515.15 (293.63)	1,464.15 (301.15)	1,410.85 (274.99)

Note: The table above reports estimates of Appendix Equation (N.6) for prime-age individuals (ages 16 to 60 years) using ACS data from 2007 to 2020. The outcomes examined are 1) an employment dummy, 2) a dummy indicating a worker is employed in a highly exposed industry, 3) usual hours worked each week, and 4) total wage and salary income. Estimates report impacts four years after the policy change ($s + 4$) relative to the year before the policy change ($s - 1$). We estimate linear probability models for binary employment outcomes such that the coefficients measure percentage point differences. For each outcome, the table presents estimates from alternative specifications with different, additive controls. Local area characteristic controls are PUMA-level density and unemployment interacted with time fixed effects and age controls include $age + age^2$. Regressions are weighted by ACS person weights and standard errors are clustered at the state-cohort level.

O. Dynamic Impacts: Minimum Wages and Imperfect Competition in Product Markets

We analyze a framework of Cournot competition with fixed costs to highlight the role of extensive margin responses and embed asymmetric production technologies to highlight how selection on productivity can further mediate the observed effects of the minimum wage. Consider a market characterized by Cournot competition among N firms facing market demand $Q_D(P)$ with asymmetric (but constant) marginal costs, c_i , which yields a familiar expression for the price-marginal cost margin.

$$\frac{P - c_i}{P} = \frac{s_i}{\varepsilon_D(Q_D)}$$

where P is the common output price, ε_D is the absolute value of the elasticity of consumer demand and varies with market quantity, and s_i denotes the market share of firm i selling quantity q_i :

$$s_i = \frac{q_i}{Q_D} = \frac{q_i}{\sum_i q_i}$$

Market shares are proportional to margins, with the most efficient firms enjoying both the largest margins and market shares. If costs are symmetric, the N firms simply split the market equally.

Minimum wages will increase marginal costs to the extent firms employ workers at wages below the new floor. As such, the marginal cost increase, Δ_i^c , can vary by firm. The resulting market share for firm i will be:

$$\frac{P^{\bar{w}} - (c_i + \Delta_i^c)}{P^{\bar{w}}} = \frac{(s_i + \Delta_i^s)}{\varepsilon_D(Q_D^{\bar{w}})} \quad (O.7)$$

where the super-script $\bar{w}$ denotes post-minimum wage hike prices and quantities, and Δ_i^s is the resulting change in market share.

Heterogeneous technology implies reallocation in response to a cost shock. When firms have heterogeneous ex-ante productivity, c_i , even when the marginal cost shock is the same for all firms, $\Delta_i^c = \Delta^c$, there will be reallocation of market shares toward the more productive firms as their margins shrink relatively less. Asymmetric cost shocks will lead to reallocation from high shock to low shock firms.⁶⁰ In any case, without a decline in the number of firms, average profits and average quantities per firm will decrease along with the total quantity in the market.

We find that minimum wage increases reduce firm entry; that is, for some firms, the added marginal cost of complying with higher wage floors will exceed market price increases, and these narrower margins will leave them unable to cover their fixed costs f , or

$$\pi_i^{\bar{w}} = (P^{\bar{w}} - (c_i + \Delta_i^c)) q_i^{\bar{w}} - f < 0.$$

Firms choosing not to enter could be those with the lowest *ex ante* productivity (high c_i), the largest cost shock due to a high expected share of low-wage labor (high Δ_i^c) per unit of

⁶⁰Depending on the differentials between the shocks, asymmetric shocks can cause profits and quantities to increase among some firms.

output, or a combination of the two. Our empirical evidence indicates that reduced entry results in a market with approximately 2% fewer firms.

Under Cournot competition, a reduction in the number of players with unchanged costs will lead to higher markups and higher market shares for operating firms, with the most cost-efficient firms gaining the most. When a cost shock reduces the number of firms, the market price will rise due to both the increase in market power and cost pass-through. Even with homogeneous firms that experience symmetric cost increases, average profits can remain flat if the number of operating firms declines sufficiently relative to the cost shock.

When heterogeneity shapes entry, such that the least productive firms or those that intended to rely most heavily on low-wage labor decide not to enter, market quantity is reallocated to more efficient firms, leading to smaller declines in market quantity and making it more likely that average profits remain unchanged. Larger markups for some firms will be matched by higher market shares ($\Delta_i^s > 0$ on the right-hand side of equation (O.7)), offsetting potentially more elastic demand as prices rise. For other firms, the added marginal cost of complying with the minimum wage can narrow margins and reduce market shares.

We find that minimum wage increases block the entry of less productive (lower value-added) and less efficient (higher material costs/revenue) firms. More efficient firms can thus benefit from minimum wage increases through the combination of higher margins and larger market shares as demand is reallocated to incumbent firms and the firms that still enter. Empirically, we estimate that average revenue increases significantly by 4.15% (0.00475) while aggregate revenue increases only 2.61% (0.0202), indicating that revenues are reallocated over a smaller number of firms following the minimum wage increase. Appendix Table F.10 shows that among incumbent firms, those with the highest baseline value-added see the strongest revenue growth. Profits do not decline among operating incumbents and entrants as a result of this added revenue.

While the model suggests that non-entry could be associated with low productivity or high exposure (i.e. a large share of low-wage labor), our findings suggest that productivity and efficiency are the main factors determining who enters. Entering firms feature high value-added and material cost efficiency, suggesting that minimum wage increases deter less productive and efficient firms, leaving the pool of actual entrants highly selected. As reported in the second column of Table 5 wage bills rise similarly among incumbent firms and entrant firms, indicating that while low-productivity firms are deterred from entry, there is no such differentiation by labor use.

The empirical entry dynamics correspond with the implications of this framework, clarifying the nature of selection and helping explain why firms seem able to weather the cost pressures of higher wage floors. By raising production costs, higher minimum wages cull the least productive independent firms and force entrants to be leaner, yielding a more productive distribution of firms after the policy change. This pattern is consistent with Sorkin (2015)’s insight that minimum wage policies shape the choices of entrants, and may hint at larger longer-run employment impacts of higher minimum wages. Through entry, and demand reallocation, the minimum wage shapes the productivity distribution of independent firms in a manner akin to how the emergence of new technologies (Collard-Wexler and De Loecker, 2015), exposure to international trade (Melitz, 2003), and recessions (Osotimehin and Pappadà, 2017) can shift the productivity distribution in an industry.

In addition to the reallocation of revenues among independent businesses, we find evi-

dence that employment relationships are reallocated from the least productive independent firms to C-corporations. Table 6 shows that low-earning workers and teens are both less likely to work at highly exposed independent businesses and more likely to work at C-corporations after minimum wage increases while Appendix Table F.10 shows that lower value-added firms drive the decline in employment relationships among incumbent independent businesses. This pattern may help explain why some corporate firms support minimum wage increases despite the higher labor costs they will face.⁶¹ Yet, we underscore that there may be other welfare-relevant margins not captured by this model. For example, Kroft et al. (2024) develop a model of imperfect competition with consumers who have love-of-variety preferences to analyze the welfare effects of commodity taxes. They show that both the externality associated with firm entry and exit, as highlighted here, and the loss of variety associated with exit are welfare-relevant and the loss of variety may dominate in many cases.

⁶¹See Amazon in addition to the polls cited in the Introduction.